AD 2. AERODROMES

VECC AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VECC - NETAJI SUBHASH CHANDRA BOSE INTERNATIONAL AIRPORT, KOLKATA

VECC AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	223914N 0882648E M. Brg. 02.32 DEG/ 1303M from physical beginning of RWY 01R	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	045 DEG/15KM from Howrah Railway Station	
3	Aerodrome elevation and reference temperature	20 FT / 36.0 DEG C	
4	Magnetic variation, date of information and annual change	0.50 DEG W (2010) /0.00	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Airports Authority of India Netaji Subash Chandra Bose International Airport, Kolkata -700052.,	
		Telephone:	(91) 33 25119944
		Fax:	
		AFS:	VECCYUYU
		Email:	apdkolkata@aai.aero
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	Geoid Model : EGM 08	

VECC AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	MON-FRI 0400-1230 UTC (0930-1800 IST) Sat, Sun and Hol : Nil
2	Custom and immigration	H24
3	Health and sanitation	H24
4	AIS briefing office	H24
5	ATS reporting office (ARO)	H24
6	MET Briefing office	H24
7	Air Traffic Service	H24
8	Fuelling	H24
9	Handling	H24
10	Security	H24
11	De-icing	NIL

12	Remarks	1. The approved hourly runway traffic handling capacity is as follows: (i) Peak capacity when runway 19L/01R in use: Maximum number of Arrival and Departure- 39 Maximum number of Arrival only – 27 Maximum number of Departure only -30 (ii) Peak capacity when runway 19R in use: Maximum no of Arrival & Departure - 29 (12 Arrivals & 11 Departures) Maximum no of Arrival only - 18 Maximum no Departure only- 30 (iii) Peak Capacity when runway 01L in use: Maximum no of Arrival & Departure - 25 (11 Arrivals & 11 Departures) Maximum no of Arrival only- 16 Maximum no Departure only – 30 2. Peak capacity during Low visibility procedures: (i) Peak capacity when RWY 19L is in use: Maximum number of Arrival & Departure: 20 (10A/10D) Maximum number of Arrival only: 15 Maximum number of Departure only: 20 (ii) Peak capacity when RWY 01R is in use: Maximum number of Arrival & Departure: 22 (11A/11D) Maximum number of Arrival only: 15 Maximum number of Departure only: 20

VECC AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	Upto 12 Tons Handling Possible.
2	Fuel and Oil types	i) Indian Oil Aviation – AVGAS 100LL and (ATF) JET A-1 fuel available ii) BPCL- ATF JET A-1 fuel available. iii) HPCL- ATF JET A-1 fuel available. iv) MWM 45.55 MJO II not in possession at AIESL, Kolkata (Centralized at AIESL, DEL) and airlines to make their own arrangements.
3	Fuelling facilities and capacity	i) Indian Oil – Refuelling rate of oil bowsers 1000 litre/min. ii) BPCL – Refuelling rate:1000 litre/min for 15 KL refuellers and 2000 litre/min for more than 23 KL refuellers. iii) HPCL – Refuelling rate: 1500 litre/min and above.
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	Available For A321/A320/A319 & B737-800 & ATR - 72 Type Of Aircraft, with AIESL. Requiring Prior Arrangements
7	Remarks	At NSCBI Airport, Kolkata, Ground Handling Service provided by: 1. Air India Airport Services Limited, Email: nonsechdule.ccu@aiasl.in; Contact: +91 9674 567 891 / +91 33 2831 8469 2. Indothai Kolkata Private Limited, E-mail: indothaiccu@indothai.in; Contact: +91 9073 964 055 / +91 33 25118100

VECC AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	Near The AD and in the city.

2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city.
3	Transportation possibilities	Taxis, Buses from AD.
4	Medical Facilities	First aid at AD. Hospitals in the city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: At AD. Open H24. Post office: At AD. Open H24.
6	Tourist office	At AD and in the City
7	Remarks	NIL

VECC AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-9
2	Rescue equipment	AVBL. as per category
3	Capability for removal of disabled aircraft	1. The critical aircraft for the airport is B777-300-ER. 2. Primary responsibility for removal of disabled aircraft rests with the concerned airline. 3. The airlines may contact 'Aerodrome Coordinator for Disabled Aircraft Removal Operations for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Airport Director, AAI, NSCBI Airport, Kolkata Contact No : 9830037890, 033-25119977 Email : apdkolkata@aai.aero
4	Remarks	The emergency access road is running parallel at a distance of 105 meter from the centreline of main runway (19L-01R) in the eastern side and connected with main runway for approaching of CFTs from Main Fire station. The road is rigid pavement having width of 9.5 meter including shoulder on both side.

VECC AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL
2	Clearance priorities	NIL
3	Remarks	NIL

VECC AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Refer Aircraft Parking/Docking Chart.Designation, surface and strength of aprons
2	Refer Aircraft Parking/Docking Chart.Designation, width, surface and strength of taxiways
3	NILLocation and elevation of altimeter checkpoints
4	At TWY A: 223941.08N 0882650.97E / ELEV: 15 FT. EGM-08Location of VOR checkpoints
5	NILPosition of INS checkpoints

6	Remarks	**1. Coordinates of RWY Holding Position**		
		TWY	**COORDINATES**	**ELEVATION (FT)**
		TWY A for RWY 01L	223811.53036N 0882633.69182E	17
		TWY A for RWY 19R	223932.16983N 0882643.06667E	19
		TWY A RWY 19R (PATTERN B)	223930.94851N 0882642.27664E	19
		TWY A RWY 19L	223941.01185N 0882651.65547E	15
		TWY B RWY 19L	223923.28964N 0882649.53508E	16
		TWY B RWY 19R	223922.47721N 0882642.01432E	18
		TWY C RWY 19L	223847.06515N 0882645.32196E	16
		TWY C RWY 19L (PATTERN B)	223847.17576N 0882644.24210E	17
		TWY C RWY 19R	223847.82287N 0882637.93000E	18
		TWY D RWY 01R	223833.28840N 0882644.13848E	16
		TWY D RWY 01R (PATTERN B)	223832.55497N 0882642.28907E	16
		TWY D RWY 01L	223829.99432N 0882635.84633E	16
		TWY E	223900.20951N 0882639.34202E	17
		TWY H	223824.37288N 0882635.13740E	16
		TWY J	223815.02174N 0882634.06935E	16
		TWY K	223957.97371N 0882653.32873E	17
		TWY P	224023.84710N 882656.48922E	16
		TWY N	223956.48785N 882645.90769E	15
		2. Geoid Model : EGM 08		

VECC AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	For Aircraft stand identification sign & SVDGS, Refer AD 2.23 Taxiway guideline marking available on all Taxiways. Appropriate signages available.

2	Runway and taxiway markings and lights	RWY19L/01R Markings: Designation, Aiming Point, THR, TDZ, Centreline, Side Strip, Chevron Marking, Runway Turn pad and Displaced THR. RWY 19L Lights: CAT-III Approach, Centreline, TDZ, THR, wing bar, RWY edge, RWY end. RWY 01R Lights: CAT-III Approach, Centreline, TDZ, THR, wing bar, RWY edge, RWY end PAPI. RWY19R/01L Markings: Designation, Aiming Point, THR, TDZ, Centreline, Side Strip, Runway Turn pad and Displaced THR. RWY 19R Lights: CAT-I Approach, THR, RWY edge, RWY end and Runway Turn pad PAPI. RWY 01L Lights: Simple Approach, THR, RWY edge and RWY end PAPI. TWY Markings: Centreline, Edge, Holding Position, Fillet. TWY Lights: Centreline lights on TWYs A, D, F, F1, F2, J1, H1, L, N1, N2, N3, N4, N, P, Q1, Q2 and R1. TWY Edge lights on all taxiways except TWY G, M, F, F1, F2, N1, N2, N3, N4, A1 and for TWY P on curves.
3	Stop bars (if any)	Available on TWY D, A, C and P
4	Remarks	Runway Guard lights have been provided for primary runway at CAT I holding positions on taxiways A, B, C, D, K and P. Intermediate Holding Position (IHP) lights have been provided on the following taxiways: i. F1 Short of junction of F1, F2, H1, J1 ii. F2 Short of junction of F1, F2, H1, J1 iii. H1 Short of junction of F1, F2, H1, J1 iv. J1 Short of junction of F1, F2, H1, J1 v. A North of junction of A & D vi. A South of junction of A & D vii. A North of junction of A, H & H1 viii. A South of junction of A & H1. ix. A Short of 19R/01L near Cat II holding position x. N1 South of junction of N & N1, N2, N3, N4 xi. N2 South of junction of N & N1, N2, N3, N4 xii. N3 South of junction of N & N1, N2, N3, N4 xiii. N4 South of junction of N & N1, N2, N3, N4 xiv. N Short of 19R GP antenna on the southern side xv. P North of junction of P & N

VECC AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19R/TKOF 01L/APCH	BUILDING	223725.2N 0882635.9E	109 FT	NIL	Big Bazaar Building
19R/TKOF 01L/APCH	BUILDING	223740.3N 0882642.4E	94 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223735.8N 0882636.1E	85 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223757.9N 0882633.1E	51 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223757.7N 0882633.9E	49 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223751.6N 0882639.6E	72 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223757.6N 0882637.3E	43 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223758.8N 0882634.9E	54 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223758.4N 0882638.3E	45 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223755.7N 0882632.8E	48 FT	NIL	BUILDING
19R/TKOF 01L/APCH	BUILDING	223750.0N 0882641.9E	72 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223744.7N 0882639.5E	82 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223745.7N 0882641.3E	68 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223741.6N 0882642.3E	71 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223745.7N 0882642.4E	65 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223744.1N 0882641.3E	67 FT	NIL	BUILDING
19R/TKOF 01L/APCH	BUILDING	223745.3N 0882640.7E	68 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223737.8N 0882642.4E	86 FT	NIL	BUILDING
19R/TKOF 01L/APCH	BRIDGE	223737.6N 0882636.5E	82 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223742.6N 0882641.9E	91 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223743.8N 0882640.3E	83 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223740.3N 0882638.3E	90 FT	NIL	BUILDING
19R/TKOF 01L/APCH	BUILDING	223744.0N 0882636.9E	80 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223742.9N 0882636.7E	82 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223736.4N 0882635.2E	86 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223752.4N 0882635.1E	55 FT	NIL	BUILDING

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19R/TKOF 01L/APCH	BUILDING	223755.6N 0882638.7E	67 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223756.7N 0882638.4E	67 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223757.0N 0882636.2E	52 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223752.7N 0882630.1E	55 FT	NIL	Building
19R/TKOF 01L/APCH	TREE	223756.8N 0882640.5E	68 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	BUILDING	223756.0N 0882631.1E	54 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223732.4N 0882632.5E	97 FT	NIL	BUILDING
19R/TKOF 01L/APCH	BUILDING	223737.1N 0882638.2E	86 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223737.4N 0882642.0E	87 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223720.4N 0882628.0E	128 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223749.4N 0882640.5E	81 FT	NIL	BUILDING
19R/TKOF 01L/APCH	BUILDING	223740.7N 0882639.0E	90 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223747.3N 0882640.8E	81 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223747.2N 0882637.6E	75 FT	NIL	Building
19R/TKOF 01L/APCH	TREE	223754.8N 0882636.4E	74 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223755.4N 0882635.8E	74 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223759.8N 0882637.1E	43 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223748.1N 0882640.9E	82 FT	NIL	Building (Ambey Garden)
19R/TKOF 01L/APCH	BUILDING	223738.8N 0882636.7E	85 FT	NIL	Building (Srijan Apartment)
19R/TKOF 01L/APCH	BUILDING	223720.0N 0882625.6E	131 FT	NIL	Building (U/C)
19R/TKOF 01L/APCH	BUILDING	223718.3N 0882627.4E	131 FT	NIL	Building (U/C)
19R/TKOF 01L/APCH	BUILDING	223759.9N 0882639.1E	43 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223759.8N 0882638.1E	46 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223759.6N 0882636.4E	45 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223759.4N 0882635.4E	45 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223759.2N 0882634.7E	53 FT	NIL	Sintex Tank on Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19R/TKOF 01L/APCH	BUILDING	223758.2N 0882633.6E	50 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223757.7N 0882633.3E	50 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223758.3N 0882633.1E	49 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223758.7N 0882632.0E	49 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223758.7N 0882631.7E	52 FT	NIL	Sintex Tank on Building
19R/TKOF 01L/APCH	BUILDING	223755.7N 0882629.4E	49 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 13.4 M.)
19R/TKOF 01L/APCH	BUILDING	223759.8N 0882638.5E	47 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 13.4 M.)
19R/TKOF 01L/APCH	BUILDING	223759.8N 0882637.7E	48 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 14.4 M.)
19R/TKOF 01L/APCH	BUILDING	223757.7N 0882631.6E	52 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 14.8 M.)
19R/TKOF 01L/APCH	BUILDING	223757.4N 0882630.2E	61 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 18.0 M.)
19R/TKOF 01L/APCH	BUILDING	223741.0N 0882639.6E	90 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 26.3 M.)
19R/TKOF 01L/APCH	BUILDING	223758.9N 0882638.4E	46 FT	NIL	Sintex Tank on Bldg. (Bldg Top 12.9 M.)
19R/TKOF 01L/APCH	TREE	223758.4N 0882633.5E	46 FT	NIL	Coconut Tree
19R/TKOF 01L/APCH	TREE	223758.2N 0882629.8E	42 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223756.8N 0882640.5E	68 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223753.3N 0882637.8E	83 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223752.6N 0882627.2E	71 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223758.1N 0882631.3E	50 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223753.0N 0882632.3E	62 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223757.7N 0882638.6E	49 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223756.4N 0882641.5E	66 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223755.6N 0882637.2E	72 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223753.4N 0882638.4E	74 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223754.2N 0882638.1E	77 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19R/TKOF 01L/APCH	TREE	223752.7N 0882636.9E	77 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223753.3N 0882636.6E	75 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223749.8N 0882636.5E	74 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223747.8N 0882636.3E	73 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223752.1N 0882635.6E	69 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223751.1N 0882633.5E	75 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223751.7N 0882631.5E	83 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223754.0N 0882629.5E	65 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223755.9N 0882630.1E	72 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223755.7N 0882634.1E	72 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223756.8N 0882630.3E	75 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223758.4N 0882634.4E	67 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223757.5N 0882636.0E	64 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223757.5N 0882636.3E	65 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223757.8N 0882640.2E	67 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223757.5N 0882628.4E	70 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223758.6N 0882630.9E	56 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223758.2N 0882632.2E	72 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223757.2N 0882633.2E	79 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223759.2N 0882633.9E	63 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223759.4N 0882635.0E	52 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223759.3N 0882636.1E	63 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223757.3N 0882640.3E	76 FT	NIL	GROUP OF TREE
19R/TKOF 01L/APCH	TREE	223748.8N 0882639.0E	96 FT	NIL	Tree
19R/TKOF 01L/APCH	TREE	223748.3N 0882636.7E	93 FT	NIL	Tree
19R/TKOF 01L/APCH	OTHER	223739.1N 0882631.6E	91 FT	NIL	Cellphone Mast

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19R/TKOF 01L/APCH	OTHER	223758.9N 0882634.2E	55 FT	NIL	Mast on Bldg.
19R/TKOF 01L/APCH	OTHER	223739.6N 0882640.5E	112 FT	NIL	Mast on Bldg. (Bldg. Top 26.8 M.)
19R/TKOF 01L/APCH	OTHER	223740.5N 0882641.1E	110 FT	NIL	Mast on Bldg. (Bldg. Top 27.3 M.)
19R/TKOF 01L/APCH	POLE	223721.6N 0882627.6E	139 FT	NIL	Pole on Building U/C
19R/TKOF 01L/APCH	OTHER	223721.5N 0882628.6E	136 FT	NIL	Pole on Building U/C (Bldg. Top 39.7 M.)
19R/TKOF 01L/APCH	SIGN	223722.8N 0882629.2E	119 FT	NIL	Advertising Board
19R/TKOF 01L/APCH	OTHER	223808.7N 0882636.5E	20 FT	NIL	Approach Light
19R/TKOF 01L/APCH	STACK	223715.9N 0882620.0E	129 FT	NIL	Chimney
19R/TKOF 01L/APCH	STACK	223741.8N 0882641.5E	81 FT	NIL	Chimney
19R/TKOF 01L/APCH	ANTENNA	223739.5N 0882637.1E	84 FT	NIL	Dish Antenna on Bldg (Bldg. Top 24.8 M.)
19R/TKOF 01L/APCH	ANTENNA	223738.4N 0882637.8E	85 FT	NIL	Dish Antenna on Bldg (Bldg. Top 25.0 M.)
19R/TKOF 01L/APCH	ANTENNA	223737.6N 0882637.6E	87 FT	NIL	Dish Antenna on Bldg (Bldg. Top 25.3 M.)
19R/TKOF 01L/APCH	ANTENNA	223739.4N 0882641.9E	89 FT	NIL	Dish Antenna on Bldg (Bldg. Top 26.8 M.)
19R/TKOF 01L/APCH	OTHER	223725.7N 0882634.5E	106 FT	NIL	L/C Rod
19R/TKOF 01L/APCH	OTHER	223801.7N 0882633.0E	34 FT	NIL	L/C Rod on Localizer Hut (Hut Top 8.9 M.)
19R/TKOF 01L/APCH	POLE	223758.7N 0882630.4E	48 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223800.1N 0882640.6E	46 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223800.2N 0882641.6E	46 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223800.2N 0882639.7E	46 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223800.3N 0882638.7E	47 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223800.3N 0882637.7E	47 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223800.1N 0882636.9E	47 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223759.8N 0882635.9E	46 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223759.6N 0882635.0E	47 FT	NIL	Lamp Post

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19R/TKOF 01L/APCH	POLE	223759.5N 0882634.1E	48 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223759.3N 0882633.2E	48 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223759.1N 0882632.3E	48 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223758.9N 0882631.3E	47 FT	NIL	Lamp Post
19R/TKOF 01L/APCH	POLE	223758.7N 0882629.4E	47 FT	NIL	Lamp Post
19R/APCH 01L/TKOF	BUILDING	224004.9N 0882649.3E	34 FT	NIL	Mosque
19R/APCH 01L/TKOF	OTHER	224003.3N 0882650.0E	28 FT	NIL	Approach Light
19R/APCH 01L/TKOF	TREE	224005.2N 0882649.2E	37 FT	NIL	Group of Trees
19R/APCH 01L/TKOF	TREE	224005.5N 0882649.1E	36 FT	NIL	Group of Trees
19R/APCH 01L/TKOF	POLE	224023.7N 0882644.7E	73 FT	NIL	Electric Pole
19R/APCH 01L/TKOF	OTHER	224004.1N 0882650.1E	31 FT	NIL	Approach Light
19R/APCH 01L/TKOF	OTHER	224005.0N 0882650.2E	34 FT	NIL	APPROACH LIGHT
19R/APCH 01L/TKOF	SIGN	224033.2N 0882643.9E	90 FT	NIL	Hoarding
19R/APCH 01L/TKOF	SIGN	223958.7N 0882652.7E	20 FT	NIL	Sign Board
19R/APCH 01L/TKOF	ANTENNA	224001.0N 0882651.8E	36 FT	NIL	Met. Antenna
19R/TKOF 01L/APCH	OTHER	223754.9N 0882637.8E	80 FT	NIL	Rod on U/C Bldg. (Bldg. Top 22.7 M.)
19R/TKOF 01L/APCH	SIGN	223808.6N 0882634.5E	20 FT	NIL	Sign Board
19L/APCH 01R/TKOF	BUILDING	224052.1N 0882658.2E	63 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224056.3N 0882704.5E	74 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224057.6N 0882706.8E	73 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223756.5N 0882639.0E	67 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223746.2N 0882638.4E	77 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223756.2N 0882631.5E	52 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	223722.9N 0882626.2E	132 FT	NIL	Building U/C
19R/TKOF 01L/APCH	BUILDING	223759.3N 0882641.1E	43 FT	NIL	Tank on Building
19L/APCH 01R/TKOF	BUILDING	224053.2N 0882710.3E	64 FT	NIL	Tank on Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19R/TKOF 01L/APCH	TREE	223800.2N 0882637.8E	52 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224103.5N 0882659.7E	95 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224106.8N 0882656.2E	95 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224058.1N 0882704.8E	81 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224053.9N 0882708.5E	85 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224056.0N 0882710.2E	87 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224059.6N 0882712.1E	87 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224111.3N 0882703.3E	100 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223755.5N 0882640.5E	79 FT	NIL	Group of Trees
19R/TKOF 01L/APCH	TREE	223749.0N 0882637.8E	77 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	OTHER	224104.1N 0882708.0E	89 FT	NIL	Mast
19R/TKOF 01L/APCH	TREE	223759.6N 0882635.5E	58 FT	NIL	Flood Light Mast
19R/TKOF 01L/APCH	OTHER	223730.6N 0882632.6E	110 FT	NIL	Mast
19R/TKOF 01L/APCH	OTHER	223751.7N 0882635.9E	71 FT	NIL	Mast on Building (Building Top 15.3m)
19R/TKOF 01L/APCH	OTHER	223754.8N 0882640.6E	63 FT	NIL	Pole on Building (Building Top 17.7m)
19L/APCH 01R/TKOF	BUILDING	224050.3N 0882701.8E	79 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224047.7N 0882656.7E	51 FT	NIL	BUILDING
19L/APCH 01R/TKOF	BUILDING	224048.9N 0882658.8E	57 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224103.0N 0882658.5E	81 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224050.6N 0882708.8E	60 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224048.5N 0882706.3E	58 FT	NIL	BUILDING
19L/APCH 01R/TKOF	BUILDING	224048.6N 0882659.8E	70 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224046.1N 0882657.2E	46 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224046.4N 0882657.1E	47 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224046.8N 0882658.3E	50 FT	NIL	Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19L/APCH 01R/TKOF	BUILDING	224050.1N 0882709.5E	57 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224050.2N 0882657.2E	55 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224048.5N 0882659.4E	72 FT	NIL	BUILDING
19L/APCH 01R/TKOF	BUILDING	224049.2N 0882700.7E	54 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224048.7N 0882706.4E	57 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224048.4N 0882706.8E	53 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224048.5N 0882706.8E	60 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224050.2N 0882708.9E	59 FT	NIL	Building
19L/APCH 01R/TKOF	BUILDING	224032.9N 0882657.4E	23 FT	NIL	Hut
19L/APCH 01R/TKOF	TANK	224100.3N 0882702.0E	101 FT	NIL	Overhead Water Tank
19L/APCH 01R/TKOF	BUILDING	224046.5N 0882657.4E	52 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 14.2 M.)
19L/APCH 01R/TKOF	BUILDING	224051.3N 0882700.4E	57 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 15.7 M.)
19L/APCH 01R/TKOF	BUILDING	224047.5N 0882657.7E	56 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 16.5 M.)
19L/APCH 01R/TKOF	BUILDING	224050.6N 0882700.0E	57 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 16.6 M.)
19L/APCH 01R/TKOF	BUILDING	224049.3N 0882703.5E	56 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 16.6 M.)
19L/APCH 01R/TKOF	BUILDING	224048.2N 0882707.0E	63 FT	NIL	Sintex Tank on Bldg.(Bldg. Top 18.5 M.)
19L/APCH 01R/TKOF	BUILDING	224049.4N 0882658.4E	66 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 18.9 M.)
19L/APCH 01R/TKOF	BUILDING	224048.5N 0882700.6E	73 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 21.5 M.)
19L/APCH 01R/TKOF	BUILDING	224048.6N 0882702.2E	62 FT	NIL	Sintex Tank on Bldg. (Dhram Priya School Top 17.3 M.)
19R/TKOF 01L/APCH	BUILDING	223759.4N 0882640.1E	41 FT	NIL	Building
19R/TKOF 01L/APCH	BUILDING	224047.9N 0882703.1E	55 FT	NIL	Building
19L/TKOF 01R/APCH	BUILDING	223733.5N 0882648.4E	134 FT	NIL	Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19L/APCH 01R/TKOF	TREE	224100.0N 0882700.4E	75 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224100.3N 0882703.8E	76 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224058.0N 0882709.5E	73 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224051.2N 0882711.2E	70 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224049.1N 0882701.6E	81 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224057.3N 0882711.1E	85 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224102.3N 0882712.6E	96 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224053.4N 0882711.4E	89 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224049.1N 0882705.0E	86 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224050.0N 0882702.9E	80 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224048.7N 0882709.5E	77 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224048.5N 0882657.2E	76 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224051.8N 0882659.3E	77 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224051.5N 0882657.5E	76 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224050.6N 0882658.6E	74 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224046.2N 0882657.4E	67 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224049.8N 0882655.6E	75 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224057.7N 0882655.5E	87 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224056.4N 0882659.2E	88 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224059.3N 0882702.7E	84 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224051.7N 0882701.2E	82 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224059.2N 0882707.3E	88 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224057.2N 0882709.6E	82 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224049.9N 0882701.5E	79 FT	NIL	GROUP OF TREE
19L/APCH 01R/TKOF	TREE	224049.0N 0882706.1E	81 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224049.2N 0882703.1E	74 FT	NIL	GROUP OF TREE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19L/APCH 01R/TKOF	TREE	224049.5N 0882700.8E	67 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224050.1N 0882656.8E	72 FT	NIL	Group of Trees
19L/APCH 01R/TKOF	TREE	224046.1N 0882656.5E	59 FT	NIL	GROUP OF TREE
19L/APCH 01R/TKOF	TREE	224053.7N 0882701.5E	96 FT	NIL	Tree
19L/APCH 01R/TKOF	TREE	224049.2N 0882704.0E	72 FT	NIL	Tree
19L/APCH 01R/TKOF	TREE	224053.3N 0882704.5E	102 FT	NIL	TREE
19L/APCH 01R/TKOF	TREE	224048.5N 0882705.4E	75 FT	NIL	Tree
19L/APCH 01R/TKOF	TREE	224107.4N 0882706.2E	103 FT	NIL	Tree
19L/APCH 01R/TKOF	TREE	224049.5N 0882706.4E	82 FT	NIL	Tree
19L/APCH 01R/TKOF	TREE	224054.0N 0882706.7E	104 FT	NIL	Tree
19L/APCH 01R/TKOF	TREE	224050.5N 0882707.6E	85 FT	NIL	Tree
19L/TKOF 01R/APCH	OTHER	223747.1N 0882649.6E	104 FT	NIL	Cellphone Mast
19L/TKOF 01R/APCH	OTHER	223745.8N 0882652.2E	110 FT	NIL	Cellphone Mast
19L/TKOF 01R/APCH	OTHER	223745.7N 0882651.0E	112 FT	NIL	Cellphone Mast
19L/APCH 01R/TKOF	POLE	224048.3N 0882708.0E	58 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224048.5N 0882710.0E	59 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224047.7N 0882704.0E	54 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224047.6N 0882702.9E	54 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224047.5N 0882701.8E	56 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224047.0N 0882659.8E	57 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224046.4N 0882658.5E	57 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224046.1N 0882657.7E	56 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224045.8N 0882656.8E	57 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224048.3N 0882708.7E	58 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224048.2N 0882707.0E	59 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224048.0N 0882705.9E	57 FT	NIL	Electric Pole

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19L/APCH 01R/TKOF	POLE	224047.8N 0882704.9E	57 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	POLE	224047.3N 0882700.8E	56 FT	NIL	Electric Pole
19L/APCH 01R/TKOF	OTHER	224048.1N 0882707.8E	78 FT	NIL	Flood Light Mast
19L/TKOF 01R/APCH	OTHER	223743.0N 0882650.9E	112 FT	NIL	Mast
19L/TKOF 01R/APCH	OTHER	223745.2N 0882650.0E	108 FT	NIL	Mast on Bldg. (Bldg. Ht. 27.0 M.)
19L/APCH 01R/TKOF	POLE	224040.5N 0882705.3E	37 FT	NIL	Pole
19L/APCH 01R/TKOF	POLE	224034.2N 0882706.1E	38 FT	NIL	Pole
19L/APCH 01R/TKOF	ANTENNA	224042.9N 0882703.5E	41 FT	NIL	RF Link Antenna
19L/APCH 01R/TKOF	ANTENNA	224046.9N 0882657.7E	66 FT	NIL	Antenna on Building
19L/APCH 01R/TKOF	STACK	224050.2N 0882711.0E	77 FT	NIL	Chimney
19L/APCH 01R/TKOF	STACK	224048.5N 0882709.1E	67 FT	NIL	Chimney
19L/APCH 01R/TKOF	NAVAID	224041.4N 0882705.6E	39 FT	NIL	GAGAN Antenna
19L/APCH 01R/TKOF	SIGN	224046.9N 0882659.1E	75 FT	NIL	Hoarding
19L/APCH 01R/TKOF	SIGN	224047.3N 0882659.8E	83 FT	NIL	Hoarding
19L/APCH 01R/TKOF	SIGN	224045.2N 0882655.5E	80 FT	NIL	Hoarding
19L/APCH 01R/TKOF	SIGN	224046.5N 0882658.5E	74 FT	NIL	Hoarding
19L/APCH 01R/TKOF	SIGN	224047.9N 0882703.7E	69 FT	NIL	Hoarding
19L/APCH 01R/TKOF	SIGN	224047.8N 0882704.0E	69 FT	NIL	Hoarding
19L/APCH 01R/TKOF	SIGN	224048.1N 0882704.7E	76 FT	NIL	Hoarding
19L/APCH 01R/TKOF	SIGN	224048.3N 0882706.4E	91 FT	NIL	Hoarding
19L/APCH 01R/TKOF	OTHER	224048.0N 0882700.5E	73 FT	NIL	L/C Rod on Bldg. (Syntex Tank Top 22.1 M. & Bldg. Top 21.2 M.)
19L/APCH 01R/TKOF	OTHER	224043.2N 0882703.6E	47 FT	NIL	L/C Rod on Localizer Hut (Hut Top.9.7 M.)
19L/APCH 01R/TKOF	POLE	224044.5N 0882655.9E	51 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224044.9N 0882656.8E	52 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224045.3N 0882657.6E	52 FT	NIL	Lamp Post

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
19L/APCH 01R/TKOF	POLE	224045.7N 0882658.4E	52 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224046.0N 0882659.3E	53 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224046.3N 0882700.1E	53 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224046.5N 0882701.0E	53 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224046.7N 0882701.9E	52 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224046.8N 0882702.8E	53 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224046.9N 0882703.7E	53 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224047.0N 0882704.6E	54 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224047.1N 0882705.5E	54 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224047.2N 0882706.4E	54 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224047.3N 0882707.5E	54 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224047.5N 0882709.3E	54 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	POLE	224047.6N 0882710.2E	54 FT	NIL	Lamp Post
19L/APCH 01R/TKOF	BUILDING	224049.0N 0882706.8E	67 FT	NIL	Shed on Bldg. (Bldg. Top 19.5 M.)
19L/APCH 01R/TKOF	SIGN	224032.9N 0882659.2E	20 FT	NIL	Sign Board
19L/APCH 01R/TKOF	SIGN	224032.7N 0882657.8E	20 FT	NIL	SIGN BOARD
19L/APCH 01R/TKOF	OTHER	224046.0N 0882700.5E	46 FT	NIL	Wooden Arch
19L/APCH 01R/TKOF	SIGN	224047.0N 0882659.9E	82 FT	NIL	Hoarding
19L/APCH 01R/TKOF	OTHER	223737.9N 0882648.9E	152 FT	NIL	LC Rod Building
19L/APCH 01R/TKOF	ANTENNA	224042.9N 0882703.5E	41 FT	NIL	RF Link Antenna
In circling area and at AD	NAVAID	223943.3N 0882643.4E	70 FT	NIL	GP Main Antenna of Rwy 19R
In circling area and at AD	NAVAID	223841.8N 0882652.2E	64 FT	NIL	GP Main Antenna of Rwy 01R
In circling area and at AD	NAVAID	224004.7N 0882702.1E	76 FT	NIL	GP Main Antenna of Rwy 19L
In circling area and at AD	NAVAID	223839.6N 0882651.9E	32 FT	NIL	GP Monitor Antenna of Rwy 01R
01L/APCH	NAVAID	224007.1N 0882702.2E	35 FT	NIL	GP Monitor Antenna of Rwy 19L

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	NAVAID	223841.9N 0882652.3E	32 FT	NIL	DME (Co-Located With Glide Path) RWY 01R
In circling area and at AD	NAVAID	223943.3N 0882643.4E	45 FT	NIL	DME (Co-Located With Glide Path) RWY 19R
In circling area and at AD	OTHER	223835.3N 0882638.3E	19 FT	NIL	PAPI
In circling area and at AD	OTHER	223845.6N 0882647.0E	20 FT	NIL	PAPI
In circling area and at AD	OTHER	224002.9N 0882658.7E	19 FT	NIL	PAPI
In circling area and at AD	NAVAID	224004.6N 0882702.3E	35 FT	NIL	DME (Co-Located With Glide Path) RWY 19L
In circling area and at AD	NAVAID	223945.8N 0882643.7E	37 FT	NIL	GP Monitor Antenna RWY 19R
In circling area and at AD	BUILDING	224004.5N 0882702.2E	28 FT	NIL	GP Hut RWY 19L
In circling area and at AD	BUILDING	224004.2N 0882701.8E	35 FT	NIL	GP Hut RWY 19L
In circling area and at AD	BUILDING	223842.1N 0882652.1E	28 FT	NIL	GP Hut of Rwy 01R
In circling area and at AD	BUILDING	223904.1N 0882654.0E	22 FT	NIL	Gun Hut
In circling area and at AD	BUILDING	223914.4N 0882648.2E	22 FT	NIL	Hut
In circling area and at AD	BUILDING	223944.9N 0882651.5E	19 FT	NIL	Hut
In circling area and at AD	BUILDING	223835.8N 0882643.9E	23 FT	NIL	Hut
In circling area and at AD	POLE	224006.9N 0882702.0E	22 FT	NIL	Pole
In circling area and at AD	POLE	223929.7N 0882649.6E	27 FT	NIL	Pole LGT.
In circling area and at AD	POLE	223843.5N 0882644.3E	30 FT	NIL	Pole Lighted
In circling area and at AD	ANTENNA	223840.4N 0882644.0E	27 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223838.4N 0882643.7E	27 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223838.0N 0882643.7E	27 FT	NIL	RVR Antenna
In circling area and at AD	OTHER	223852.6N 0882653.9E	39 FT	NIL	WDI
In circling area and at AD	OTHER	223943.8N 0882651.9E	30 FT	NIL	Mobile Road Traffic (Road Elev. 4.3 M. + Traffic Ht. 5.0 M.)
In circling area and at AD	OTHER	224030.2N 0882704.4E	30 FT	NIL	Mobile Road Traffic (Road Elev. 4.0 M. + Traffic Ht. 5.0 M.)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	223807.5N 0882648.0E	29 FT	NIL	Mobile Road Traffic (Road Elev. 3.9 M. + Traffic Ht. 5.0 M.)
In circling area and at AD	OTHER	223945.8N 0882651.4E	19 FT	NIL	Blue Box
In circling area and at AD	OTHER	224003.2N 0882654.1E	22 FT	NIL	Blue Box
In circling area and at AD	OTHER	224000.9N 0882654.0E	21 FT	NIL	Blue Box
In circling area and at AD	OTHER	223828.9N 0882637.7E	22 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223828.3N 0882637.6E	22 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223827.6N 0882637.5E	22 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223826.9N 0882637.4E	22 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223827.5N 0882640.7E	21 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223826.9N 0882641.1E	21 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223826.2N 0882641.1E	20 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223825.5N 0882640.9E	21 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223825.0N 0882640.4E	21 FT	NIL	Dumble Lights
In circling area and at AD	OTHER	223842.3N 0882652.5E	21 FT	NIL	Electric Box
In circling area and at AD	OTHER	223842.3N 0882652.5E	23 FT	NIL	Electric Box
In circling area and at AD	OTHER	223840.5N 0882644.0E	24 FT	NIL	Electric Box
In circling area and at AD	OTHER	223841.6N 0882644.4E	23 FT	NIL	Electric Box
In circling area and at AD	OTHER	223918.9N 0882648.4E	22 FT	NIL	Electric Box
In circling area and at AD	OTHER	223929.8N 0882649.5E	20 FT	NIL	Electric Box
In circling area and at AD	OTHER	223930.3N 0882648.7E	17 FT	NIL	Electric Box
In circling area and at AD	OTHER	223930.5N 0882650.6E	17 FT	NIL	Electric Box
In circling area and at AD	OTHER	223930.7N 0882650.7E	17 FT	NIL	Electric Box
In circling area and at AD	OTHER	223931.3N 0882650.8E	17 FT	NIL	Electric Box
In circling area and at AD	OTHER	223931.6N 0882650.9E	17 FT	NIL	Electric Box
In circling area and at AD	OTHER	223932.2N 0882649.5E	21 FT	NIL	Electric Box
In circling area and at AD	OTHER	224021.2N 0882655.7E	26 FT	NIL	L/C Rod

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	223841.2N 0882644.0E	28 FT	NIL	L/C Rod
In circling area and at AD	OTHER	223841.4N 0882644.5E	28 FT	NIL	L/C Rod
In circling area and at AD	OTHER	223837.8N 0882643.7E	40 FT	NIL	L/C Rod
In circling area and at AD	OTHER	223918.5N 0882648.6E	26 FT	NIL	L/C Rod
In circling area and at AD	OTHER	223943.2N 0882643.5E	35 FT	NIL	L/C Rod on GP Hut (Hut Top 8.7 M.) Base 5.2 M.
In circling area and at AD	OTHER	224005.6N 0882654.4E	29 FT	NIL	L/C Rod On RVR Antenna (RVR Ant. Top 8.0 M.)
In circling area and at AD	POLE	223932.4N 0882649.9E	26 FT	NIL	Lamp Post
In circling area and at AD	POLE	224005.8N 0882654.5E	27 FT	NIL	Pole Lighted
In circling area and at AD	OTHER	223900.5N 0882646.0E	19 FT	NIL	Landing T Light
In circling area and at AD	ANTENNA	223837.9N 0882643.6E	41 FT	NIL	Met. Antenna
In circling area and at AD	SIGN	223903.8N 0882649.1E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223902.7N 0882641.6E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223909.4N 0882640.5E	22 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223932.3N 0882644.2E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223927.5N 0882642.6E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223923.9N 0882642.1E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223921.7N 0882642.2E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223922.0N 0882643.0E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223922.4N 0882643.6E	22 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223920.4N 0882643.6E	22 FT	NIL	Sign Board
In circling area and at AD	SIGN	223914.3N 0882648.1E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223933.7N 0882645.3E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223932.8N 0882642.5E	22 FT	NIL	Sign Board
In circling area and at AD	SIGN	223931.9N 0882644.0E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223931.4N 0882641.5E	21 FT	NIL	Sign Board

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	SIGN	223930.6N 0882643.2E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223926.6N 0882644.4E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223919.9N 0882641.7E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223924.5N 0882646.8E	22 FT	NIL	Sign Board
In circling area and at AD	SIGN	223943.2N 0882644.8E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223942.1N 0882651.9E	16 FT	NIL	Sign Board
In circling area and at AD	SIGN	223941.9N 0882652.4E	16 FT	NIL	Sign Board
In circling area and at AD	SIGN	224029.7N 0882657.4E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223833.8N 0882638.2E	27 FT	NIL	Sign Board
In circling area and at AD	SIGN	223845.9N 0882637.7E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223846.9N 0882637.9E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223846.9N 0882638.3E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223846.7N 0882639.7E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223848.9N 0882638.6E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223849.8N 0882638.2E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223848.5N 0882639.6E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223848.8N 0882639.9E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223837.1N 0882646.1E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223831.7N 0882642.8E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223831.2N 0882641.5E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223833.6N 0882641.9E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223833.5N 0882641.7E	19 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223829.3N 0882640.2E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223833.2N 0882641.0E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223825.9N 0882636.7E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223826.8N 0882637.3E	20 FT	NIL	Sign Board

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	SIGN	223829.4N 0882637.1E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223829.2N 0882636.3E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223826.3N 0882635.4E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223825.9N 0882635.3E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223823.2N 0882635.4E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223823.3N 0882635.6E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223854.2N 0882647.1E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223857.3N 0882641.0E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223848.4N 0882643.6E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223846.3N 0882643.4E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223846.2N 0882644.2E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223852.3N 0882643.1E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223852.0N 0882643.3E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223851.7N 0882643.1E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223851.0N 0882644.9E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223850.7N 0882644.7E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223850.4N 0882644.4E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223849.8N 0882647.6E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223844.7N 0882647.1E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223849.9N 0882650.1E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	223841.9N 0882651.3E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223841.7N 0882652.8E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223844.5N 0882649.5E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223845.1N 0882651.5E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223847.6N 0882651.8E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223923.5N 0882642.2E	21 FT	NIL	Sign Board

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	SIGN	223921.0N 0882646.2E	21 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223942.1N 0882649.8E	18 FT	NIL	Sign Board
In circling area and at AD	SIGN	223939.9N 0882651.6E	17 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223937.7N 0882653.2E	18 FT	NIL	Sign Board
In circling area and at AD	SIGN	223925.6N 0882651.8E	18 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223924.2N 0882649.7E	18 FT	NIL	Sign Board
In circling area and at AD	SIGN	223922.3N 0882649.5E	17 FT	NIL	Sign Board
In circling area and at AD	SIGN	223920.4N 0882651.2E	18 FT	NIL	Sign Board
In circling area and at AD	SIGN	224004.0N 0882656.2E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	224004.8N 0882701.2E	21 FT	NIL	Sign Board
In circling area and at AD	SIGN	224007.5N 0882701.3E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223834.2N 0882643.7E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223831.9N 0882644.5E	20 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223826.9N 0882643.2E	19 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223816.1N 0882634.4E	19 FT	NIL	SIGN BOARD
In circling area and at AD	SIGN	223813.3N 0882635.6E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223814.0N 0882634.1E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223812.8N 0882633.9E	18 FT	NIL	Sign Board
In circling area and at AD	OTHER	223933.1N 0882648.5E	18 FT	NIL	ELECTRIC BOX
In circling area and at AD	SIGN	223940.0N 0882649.6E	18 FT	NIL	Sign Board
In circling area and at AD	OTHER	223942.7N 0882648.9E	18 FT	NIL	PAPI
In circling area and at AD	OTHER	224005.7N 0882639.1E	99 FT	NIL	Cellphone Mast
In circling area and at AD	TREE	224018.0N 0882641.8E	80 FT	NIL	Group of Tree
In circling area and at AD	SIGN	223810.6N 0882633.6E	20 FT	NIL	Sign Board
In circling area and at AD	POLE	223916.2N 0882642.4E	24 FT	NIL	Signal Pole
In circling area and at AD	OTHER	223919.2N 0882648.5E	21 FT	NIL	White Box

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	OTHER	224021.6N 0882656.0E	21 FT	NIL	White Box
In circling area and at AD	OTHER	223838.7N 0882643.8E	21 FT	NIL	White Box
In circling area and at AD	OTHER	223837.8N 0882643.8E	23 FT	NIL	White Box
In circling area and at AD	ANTENNA	223918.1N 0882648.4E	25 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223919.0N 0882648.5E	25 FT	NIL	RVR Antenna
In circling area and at AD	BUILDING	224005.4N 0882654.0E	23 FT	NIL	Gun Hut
In circling area and at AD	ANTENNA	223945.9N 0882651.5E	22 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223946.9N 0882651.6E	22 FT	NIL	RVR Antenna
In circling area and at AD	BUILDING	223927.9N 0882649.8E	21 FT	NIL	GUN HUT
In circling area and at AD	ANTENNA	224003.1N 0882654.1E	27 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	224005.2N 0882654.4E	27 FT	NIL	RVR ANTENNA
In circling area and at AD	ANTENNA	224020.7N 0882655.8E	25 FT	NIL	RVR ANTENNA
In circling area and at AD	ANTENNA	224021.7N 0882655.9E	25 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223840.4N 0882644.0E	27 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223838.4N 0882643.7E	27 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223838.0N 0882643.7E	27 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223932.2N 0882650.0E	28 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223931.8N 0882650.0E	26 FT	NIL	RVR ANTENNA
In circling area and at AD	ANTENNA	223929.8N 0882649.7E	26 FT	NIL	RVR Antenna
In circling area and at AD	OTHER	224004.4N 0882702.4E	32 FT	NIL	Lightning Arrester on GP Hut (Hut Top 8.7m)
In circling area and at AD	BUILDING	223849.0N 0882645.7E	21 FT	NIL	Security Hut
In circling area and at AD	BUILDING	223900.5N 0882646.6E	21 FT	NIL	Security Hut
In circling area and at AD	ANTENNA	224020.7N 0882655.8E	25 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	224021.7N 0882655.9E	25 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223840.4N 0882644.0E	27 FT	NIL	RVR Antenna
In circling area and at AD	ANTENNA	223838.4N 0882643.7E	27 FT	NIL	RVR Antenna

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	ANTENNA	223838.0N 0882643.7E	27 FT	NIL	RVR Antenna
In circling area and at AD	BUILDING	223804.4N 0882627.6E	70 FT	NIL	Building
In circling area and at AD	BUILDING	224001.9N 0882638.6E	115 FT	NIL	Building
In circling area and at AD	BUILDING	223739.7N 0882644.0E	105 FT	NIL	Building
In circling area and at AD	BUILDING	223738.9N 0882643.6E	101 FT	NIL	Building
In circling area and at AD	BUILDING	223810.4N 0882627.2E	89 FT	NIL	Building
In circling area and at AD	BUILDING	223812.7N 0882627.1E	85 FT	NIL	Building
In circling area and at AD	BUILDING	223757.0N 0882625.6E	94 FT	NIL	Building
In circling area and at AD	BUILDING	223801.8N 0882627.5E	95 FT	NIL	Building
In circling area and at AD	BUILDING	223802.7N 0882627.5E	95 FT	NIL	Building
In circling area and at AD	BUILDING	223804.8N 0882627.3E	81 FT	NIL	Building
In circling area and at AD	BUILDING	223739.4N 0882645.0E	115 FT	NIL	Sintex Tank on Bldg. (Bldg. Top 32.4 M.)
In circling area and at AD	BUILDING	223748.6N 0882642.4E	79 FT	NIL	Building
In circling area and at AD	BUILDING	223802.7N 0882626.5E	98 FT	NIL	Building
In circling area and at AD	TREE	223758.2N 0882642.5E	64 FT	NIL	Group of Trees
In circling area and at AD	TREE	223756.5N 0882644.1E	78 FT	NIL	Group of Trees
In circling area and at AD	TREE	224007.0N 0882643.1E	82 FT	NIL	Group of Trees
In circling area and at AD	TREE	224004.2N 0882642.7E	62 FT	NIL	Group of Trees
In circling area and at AD	TREE	224010.7N 0882641.0E	84 FT	NIL	Group of Trees
In circling area and at AD	TREE	223752.6N 0882644.0E	81 FT	NIL	Group of Trees
In circling area and at AD	TREE	223756.4N 0882642.3E	71 FT	NIL	Group of Trees
In circling area and at AD	TREE	223756.0N 0882643.3E	72 FT	NIL	Group of Trees
In circling area and at AD	TREE	223810.0N 0882628.5E	70 FT	NIL	Group of Trees
In circling area and at AD	TREE	223800.4N 0882627.6E	64 FT	NIL	Group of Trees
In circling area and at AD	TREE	223757.8N 0882627.2E	58 FT	NIL	Group of Trees
In circling area and at AD	TREE	224008.8N 0882642.9E	74 FT	NIL	Tree

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TREE	224007.9N 0882640.7E	95 FT	NIL	Tree
In circling area and at AD	TREE	224008.0N 0882642.7E	86 FT	NIL	Tree
In circling area and at AD	TREE	224006.5N 0882642.7E	81 FT	NIL	Tree
In circling area and at AD	TREE	223753.4N 0882642.8E	75 FT	NIL	Tree
In circling area and at AD	TREE	223812.9N 0882628.9E	67 FT	NIL	Tree
In circling area and at AD	TREE	224039.1N 0882710.5E	71 FT	NIL	Group of Trees
In circling area and at AD	TREE	224044.8N 0882653.6E	69 FT	NIL	Group of Trees
In circling area and at AD	TREE	223758.5N 0882627.4E	65 FT	NIL	Group of Trees
In circling area and at AD	OTHER	223738.7N 0882644.5E	118 FT	NIL	Cellphone Mast on Bldg. (Bldg. Top 32.8 M.)
In circling area and at AD	OTHER	224025.7N 0882643.2E	78 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	223817.2N 0882627.0E	94 FT	NIL	Mast
In circling area and at AD	OTHER	223844.9N 0882633.5E	52 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	223838.5N 0882632.7E	52 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	223837.0N 0882632.5E	52 FT	NIL	Flood Light Mast
In circling area and at AD	FENCE	223807.9N 0882630.9E	28 FT	NIL	Airport Boundary Wall With Fencing on Top
In circling area and at AD	OTHER	223809.1N 0882631.2E	30 FT	NIL	Mobile Road Traffic (Road Elev.4.2 M. + Traffic Ht. 5.0 M.)
In circling area and at AD	ANTENNA	223818.6N 0882642.8E	31 FT	NIL	Antenna on Localizer Hut (Hut Top 7.3 M.)
In circling area and at AD	STACK	223812.7N 0882626.7E	91 FT	NIL	Cement Tank on Bldg
In circling area and at AD	STACK	223813.5N 0882628.2E	72 FT	NIL	Chimney
In circling area and at AD	OTHER	223818.4N 0882642.8E	33 FT	NIL	L/C Rod On Hut (Localizer Hut Top 7.3 M.)
In circling area and at AD	POLE	223810.0N 0882629.9E	46 FT	NIL	Lamp Post
In circling area and at AD	POLE	223800.3N 0882642.5E	46 FT	NIL	Lamp Post
In circling area and at AD	POLE	223809.1N 0882631.0E	27 FT	NIL	Lamp Post
In circling area and at AD	POLE	223808.4N 0882630.9E	26 FT	NIL	Lamp Post

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	POLE	223759.2N 0882628.6E	47 FT	NIL	Lamp Post
In circling area and at AD	POLE	223805.9N 0882629.2E	47 FT	NIL	Lamp Post
In circling area and at AD	POLE	223806.7N 0882629.7E	47 FT	NIL	LAMP POST
In circling area and at AD	POLE	223809.2N 0882630.4E	47 FT	NIL	Lamp Post
In circling area and at AD	POLE	223808.3N 0882630.5E	47 FT	NIL	Lamp Post
In circling area and at AD	POLE	223807.5N 0882630.2E	47 FT	NIL	Lamp Post
In circling area and at AD	SIGN	223943.6N 0882653.9E	17 FT	NIL	Sign Board
In circling area and at AD	SIGN	223955.5N 0882655.2E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223957.0N 0882654.2E	20 FT	NIL	Sign Board
In circling area and at AD	SIGN	223941.9N 0882653.0E	16 FT	NIL	Sign Board
In circling area and at AD	SIGN	224030.7N 0882704.2E	19 FT	NIL	Sign Board
In circling area and at AD	SIGN	223924.1N 0882650.9E	17 FT	NIL	Sign Board
In circling area and at AD	SIGN	223843.5N 0882649.3E	21 FT	NIL	Sign Board
In circling area and at AD	ANTENNA	223949.4N 0882638.6E	87 FT	NIL	SMR Antenna
In circling area and at AD	BUILDING	224044.6N 0882654.0E	64 FT	NIL	Building
In circling area and at AD	BUILDING	224045.0N 0882654.9E	49 FT	NIL	Building
In circling area and at AD	BUILDING	223758.8N 0882641.9E	44 FT	NIL	Building
In circling area and at AD	TREE	224052.8N 0882653.6E	90 FT	NIL	GROUP OF TREE
In circling area and at AD	TREE	224104.9N 0882715.8E	100 FT	NIL	GROUP OF TREE
In circling area and at AD	TREE	224058.1N 0882713.7E	89 FT	NIL	Group of Trees
In circling area and at AD	TREE	224053.6N 0882713.3E	101 FT	NIL	Group of Trees
In circling area and at AD	TREE	224045.9N 0882653.5E	76 FT	NIL	Group of Trees
In circling area and at AD	TREE	224052.2N 0882654.6E	86 FT	NIL	Group of Trees
In circling area and at AD	TREE	224026.0N 0882709.1E	77 FT	NIL	Group of Trees
In circling area and at AD	TREE	224045.5N 0882654.1E	78 FT	NIL	Tree
In circling area and at AD	TREE	224045.9N 0882652.8E	79 FT	NIL	Tree

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TREE	223945.6N 0882705.0E	82 FT	NIL	Group of Trees
In circling area and at AD	POLE	224044.9N 0882655.0E	55 FT	NIL	Electric Pole
In circling area and at AD	OTHER	223925.5N 0882634.6E	119 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	224013.3N 0882653.5E	38 FT	NIL	WDI
In circling area and at AD	OTHER	223851.3N 0882634.4E	52 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	223852.2N 0882634.5E	52 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	223855.2N 0882634.9E	52 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	223855.7N 0882635.0E	52 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	223829.0N 0882651.8E	29 FT	NIL	Mobile Road Traffic (Road Elev.3.8 M. + Traffic Ht. 5.0 M.)
In circling area and at AD	ANTENNA	223917.9N 0882657.8E	29 FT	NIL	Antenna
In circling area and at AD	SIGN	224046.5N 0882711.8E	78 FT	NIL	Hoarding
In circling area and at AD	SIGN	224044.9N 0882654.5E	90 FT	NIL	Hoarding
In circling area and at AD	SIGN	224045.3N 0882655.2E	80 FT	NIL	Hoarding
In circling area and at AD	SIGN	224044.3N 0882653.3E	90 FT	NIL	Hoarding
In circling area and at AD	SIGN	224046.4N 0882711.6E	77 FT	NIL	Hoarding
In circling area and at AD	OTHER	223926.3N 0882702.7E	83 FT	NIL	L/C Rod on MSSR
In circling area and at AD	POLE	224044.2N 0882655.1E	51 FT	NIL	Lamp Post
In circling area and at AD	BUILDING	224046.9N 0882655.0E	58 FT	NIL	Shed on Bldg. (Bldg. Top 14.6 M.)
In circling area and at AD	SIGN	223917.4N 0882639.6E	22 FT	NIL	Sign Board
In circling area and at AD	SIGN	223940.0N 0882649.6E	18 FT	NIL	Sign Board
In circling area and at AD	SIGN	224003.2N 0882642.0E	58 FT	NIL	Hoarding
In circling area and at AD	BUILDING	223719.0N 0882703.7E	172 FT	NIL	Building
In circling area and at AD	BUILDING	223718.4N 0882706.8E	169 FT	NIL	Greenwood Sonata Building
In circling area and at AD	BUILDING	223705.4N 0882748.0E	205 FT	NIL	Building U/C
In circling area and at AD	ANTENNA	223720.3N 0882659.4E	180 FT	NIL	Antenna on Swissotel (Bldg. Top 51.0 M.)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	BUILDING	223720.6N 0882804.8E	168 FT	NIL	Building
In circling area and at AD	BUILDING	223715.4N 0882807.3E	172 FT	NIL	Building (Hiland Woods Apartment)
In circling area and at AD	BUILDING	223900.9N 0882622.8E	188 FT	NIL	New ATC Building U/C
In circling area and at AD	BUILDING	223717.7N 0882805.5E	279 FT	NIL	Building (Akanksha)
In circling area and at AD	BUILDING	224052.4N 0882745.3E	169 FT	NIL	Building
In circling area and at AD	OTHER	223718.1N 0882714.2E	173 FT	NIL	LC Rod Building (Building Top 49.0m)
In circling area and at AD	TOWER	222943.1N 0882139.1E	613 FT	NIL	TV Tower
In circling area and at AD	OTHER	223254.1N 0882059.0E	850 FT	NIL	Lightning Arrester on Building (Building Top 255.1m)

VECC AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	AMO Kolkata
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	H24
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	Kolkata 9, 30HR
4	Availability of the trend forecast for the aerodrome and interval of issuance	Trend Trend Type landing forecast appended with each METAR / SPECI
5	Information on how briefing and/or consultation is provided	Provided, , through Documentation and OLBS
6	Types of flight documentation supplied and language(s) used in flight documentation	Chart And Tabular English
7	Charts and other information displayed or available for briefing or consultation	1. WAFC Products: Data for upper wind & temperature for levels: 850 h.Pa, 700 h.Pa, 500 h.Pa, 300 h.Pa, 250 h.Pa, 200 h.Pa, 150 h.Pa, SIG Wx charts for FL 100 - 250 & for FL 250 - 630. 2. National Charts for consultation only. Surface charts, synoptic upper air charts for levels: 100, 150, 200, 250, 300, 400, 500, 600, 700, 850 hPa, prognostic upper air charts at constant pressure, prognostic significant weather chart, DWR products / Satellite imagery.
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	RS / RW Ascent, Transmissometer at TDZ 19L, TDZ 19R, MID 19L, TDZ 01R, MID 01R, END 19L & END 01R, AWOS at TDZ 19L & TDZ 01R, Radar & Satellite Picture
9	The air traffic services unit(s) provided with meteorological information	Kolkata ATC, ATS and ACC
10	Additional information, e.g. concerning any limitation of service.	Ceilometer at Runway 01R End operational .

VECC AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCR) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
01L	6.20 DEG	3271 x 45 M	786/F/B/W/T Asphalt	THR: 223824.95N 0882638.43E RWY END: 223956.50N 0882649.16E
19R	186.20 DEG	3271 x 45 M	786/F/B/W/T Asphalt	THR: 223953.65N 0882648.82E RWY END: 223810.79N 0882636.78E
01R	6.20 DEG	3633 x 45 M	721/F/A/W/T Asphalt	THR: 223831.95N 0882646.76E RWY END: 224029.36N 0882700.51E
19L	186.20 DEG	3633 x 45 M	721/F/A/W/T Asphalt	THR: 224015.57N 0882658.89E RWY END: 223831.95N 0882646.76E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 18.0FT TDZ: 19.0FT		NIL	NIL	3391 x 266 M
THR: 17.0FT TDZ: 19.0FT		NIL	NIL	3391 x 266 M
THR: 18.0FT TDZ: 19.0FT		NIL	NIL	3753 x 300 M
THR: 18.0FT TDZ: 19.0FT		NIL	NIL	3753 x 300 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
165M x 90M	NIL	AVBL	1. PCR (CH: 0M – 438M): 1025/R/B/W/T, Surface: concrete; PCR (CH: 438M – 750M): 1140/R/B/W/T, Surface: concrete; PCR (CH:750M – 3271M):786/F/B/W/T, Surface: Bitumen. 2. Slope: -0.33% (0 to 60M), +0.14% (60m to 360m), -0.17% (420m to 780m), +0.1% (780m to 1380m), -0.04% (1380m to 1830m), +0.14% (1830m to 2190m), - 0.07% (2190m to 3271m) 3. Geoid Model : EGM 08
220M x 90M		AVBL	1. Overall RWY Slope: (-)0.006% 2. Geoid Model: EGM 08
240M x 90M		AVBL	1.PCR (CH: 0M – 3573M) 721/F/A/W/T, Surface: Bitumen; PCR (CH: 3573M – 3627M): 984/R/A/W/T, Surface: concrete. 2. Slope: +0.12% (0 to 330m), -0.07% (330m to 2100m), +0.15% (2100m to 2850m), -0.1% (2850m to 3633m) 3. Geoid Model: EGM 08
240M x 90M		AVBL	1. Overall RWY Slope: (-) 0.01% 2. Geoid Model: EGM 08

VECC AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
01L	3271	3271	3271	2833	RWY displaced by 438M.
19R	3271	3271	3271	3183	RWY displaced by 88M.
01R	3633	3633	3633	3633	
19L	3633	3633	3633	3207	RWY displaced by 426M.

VECC AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
01L	SALS 420 M LIH	Green Wing Bar - NIL	PAPI LEFT/3.00 DEG MEHT (50.00FT)	
19R	CAT I 900 M LIH	Green Wing Bar - NIL	PAPI LEFT/3.00 DEG MEHT (50.00FT)	
01R	CAT III 900 M LIH	Green Wing Bar- Available	PAPI LEFT/3.00 DEG MEHT (50.00FT)	900 M
19L	CAT III 900 M LIH	Green Wing Bar- Available	PAPI LEFT/3.00 DEG MEHT (50.00FT)	900 M

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
	3271 M 60 M Variable white LIH	Red		1. APCH with cross bar at 300M.
	3271 M 60 M Variable white LIH	Red		1. Last 600M of runway edge lights amber.
3633 M 15 M LIHVariable White	3633 M 60 M Variable white LIH	Red		1. Runway Centreline lights Last 900m to 300m from RWY end alternate red & variable white Red for last 300m from RWY end 2. Last 600M of Runway Edge lights amber.
3633 M 15 M LIHVariable White	3633 M 60 M Variable white LIH	Red		1. RWY Centreline lightsLast 900m to 300m from RWY end alternate red & variable white Red for last 300m from RWY end 2. Last 600M ofrunway edge lights amber. 3. Inset Turn Pad Lights on Runway 19L Available InSingle Circuit with Turn Pad Edge Light

VECC AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	Installed on the top of ATC tower; Beam colour- Green & White; 24-30 flashes per minutes; H-24 H24
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	Available
		Anemometer	1. 390 M from THR19L, Lighted 2. 300 M from THR01R, Lighted (Both 1&2 located North of RWY 19L / 01R)
3	Taxiway edge and taxiway centre line lights;	Edge	All Taxiways except TWY-G, M, F, F1, F2, N1, N2, N3, N4, A1 and P on curves.
		Centre Line	ON TWYs A,D, F, F1, F2, J1,H1, L, N1, N2, N3, N4, N, P, Q1, Q2 and R1
4	Secondary power supply including switch-over time;	Secondary power supply to all air field lighting. Switch-over time : CAT-I 15 SEC CATII/III 1 sec.	
5	Remarks		

VECC AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not Established
2	TLOF and/or FATO area elevation:	Not Established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not Established
4	True bearings of FATO;	Not Established
5	Declared distances available	Not Established
6	Approach and FATO lighting;	Not Established
7	Remarks	Not Established

VECC AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 223914N 0882648E within a 25NM radius.
2	Vertical limits	FL 50
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Kolkata Tower / Kolkata Radar, English
5	Transition altitude	4000 FT
6	Hours of applicability	H24
7	Remarks	Airspace from 4000 FT AMSL and above is Classified as Class ‘C’.

VECC AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	2	3	4
OAC	KOLKATA CONTROL	123.775 MHZ	SHORT CODE : 441921 PSTN: 00918991117566 SATVOICE AVBL WITH WSO: SHORT CODE : 441902 PSTN: 00913325119520
OAC	KOLKATA CONTROL	128.300 MHZ	SHORT CODE : 441921 PSTN: 00918991117566 SATVOICE AVBL WITH WSO: SHORT CODE : 441902 PSTN: 00913325119520
OTHER	Kolkata Delivery	121.625 MHZ	
FIS	Kolkata Information	127.175 MHZ	
SAR	-	123.100 MHZ	SHORT CODE : 441932 PSTN NUMBER : 00918991117567
APP	Kolkata Radar	119.300 MHZ	
APP	Kolkata Radar	120.025 MHZ	
APP	Kolkata Radar	120.675 MHZ	
APP	Kolkata Radar	126.075 MHZ	
APP	Kolkata Radar	127.300 MHZ	
APP	Kolkata Radar	127.900 MHZ	
TWR	Kolkata Tower	118.025 MHZ	
TWR	Kolkata Tower	118.100 MHZ	
ATIS	Kolkata Information	126.400 MHZ	
ACC	Kolkata Radar	120.100 MHZ	
ACC	Kolkata Radar	120.575 MHZ	
ACC	Kolkata Radar	120.700 MHZ	
ACC	Kolkata Radar	124.650 MHZ	
ACC	Kolkata Radar	125.775 MHZ	
ACC	Kolkata Radar	125.900 MHZ	
ACC	Kolkata Radar	126.100 MHZ	
ACC	Kolkata Radar	127.525 MHZ	
ACC	Kolkata Radar	132.250 MHZ	
ACC	Kolkata Radar	132.350 MHZ	
ACC	Kolkata Radar	133.750 MHZ	
ACC	Kolkata Radar	134.375 MHZ	
ALRS	---------	121.500 MHZ	

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
SMC	Kolkata Ground	121.725 MHZ	
SMC	Kolkata Ground	121.900 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	**6**	**7**
	H24	1. KOLKATA OCEANIC AREA CONTROL CENTRE, NORTH 2. MAIN FREQ 3. SATVOICE AVBL WITH WSO
	H24	1. KOLKATA OCEANIC AREA CONTROL CENTRE, SOUTH 2. MAIN FREQ 3. SATVOICE AVBL WITH WSO
	H24	Clearance Delivery
		1. FREQUENCY 127.175MHZ AVAILABLE AROUND 'CEA' DVOR. 2. FOR OTHER AREAS, RESPECTIVE SECTOR FREQUENCY AS MENTIONED IN EAIP INDIA VER2.0 ENR2.1 TO BE USED FOR FLIGHT INFORMATION SERVICE.
	H24	International Voice SAR On Scene Frequency 123.1MHZ Is Operational
	H24	1. Approach Departure 2. Approach Procedural 3. Backup Frequency
	H24	1. APPROACH SEQUENCE 2. BACKUP FREQUENCY.
	H24	APPROACH ARRIVAL MAIN FREQ.
	H24	APPROACH ARRIVAL SDBY FREQ.
	H24	Approach Sequence
	H24	1. Approach Departure 2. Approach Procedural 3. Main Frequency
	H24	SDBY FREQ
	H24	Main Frequency
	H24	NIL
	H24	1. Lower Kolkata West 2. Main Frequency 3. Backup Frequency 132.250 MHz
	H24	1. Upper Kolkata East 2. Main Frequency 3. Backup Frequency 120.700 MHz
	H24	1. Lower Kolkata East 2. Main Frequency 3. Backup Frequency 120.575 MHz
	H24	1. Upper Varanasi 2. Backup Frequency
	H24	1. Upper Guwahati
	H24	1. Upper Raipur
	H24	1. Lower Kolkata North 2. Main Frequency 3. Backup Frequency 133.750 MHz

Logon address, as appropriate	Hours of operation	Remarks
	H24	1. Upper Varanasi 2. Main Frequency
	H24	1. Upper Kolkata West 2. Main Frequency 3. Backup Frequency 120.100 MHz
	H24	1. Upper Bhubaneswar 2. Main Frequency 3. Backup Frequency 132.450 MHz
	H24	1. Upper Kolkata North 2. Main Frequency 3. Backup Frequency 126.100 MHz
	H24	1-Upper Raipur 2-Backup FREQ
	H24	EMERGENCY FREQUENCY
	H24	SDBY FREQ
	H24	MAIN FREQ.

VECC AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 01R CAT III	ICAL	109.900 MHz	H24
LOC 19L CAT III	IDUM	110.300 MHz	H24
LOC 19R	IOKL	111.300 MHz	H24
GP 01R		333.800 MHz	H24
GP 19L		335.000 MHz	H24
GP 19R		332.300 MHz	H24
DME ILS 01R	ICAL	CH36X	H24
DME ILS 19L	IDUM	CH40X	H24
DME ILS 19R	IOKL	CH50X	H24
DVOR/DME	CEA	112.500 MHz CH72X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
224040.8N 0882701.9E			
223818.2N 0882645.2E			
223801.6N 0882635.7E			LLZ inner coverage is restricted to 30 DEG on 150HZ side, no restriction on 90HZ side.
223841.8N 0882652.2E			3 DEG
224004.6N 0882701.9E			Glide angle 3 DEG
223943.3N 0882643.4E			3 DEG
223841.9N 0882652.3E	32 FT		1. Collocated with GP 01R 2. Geoid Model EGM08
224004.6N 0882702.1E	37 FT		
223943.3N 0882643.4E	45 FT		1. Collocated with GP 19R 2. Geoid Model- EGM08
224035.4N 0882652.5E	47 FT		(I) Gravity Model: EGM-08 (II) DME Colocated with DVOR.

VECC AD 2.20 LOCAL AERODROME REGULATIONS

I.AIRPORT COLLABORATIVE DECISION MAKING (A-CDM, KOLKATA)

1. BACKGROUND

1.1 The NSCBI Airport, Kolkata - Collaborative Decision Making (A-CDM, Kolkata) undertaken at NSCBI Airport is a joint programme among all airport partners –

- Air Navigation Service Provider (ATC)
- Aircraft Operators (AO)
- Airport Operator (AOCC)
- Ground Handling Agencies (GHA)
- Central Air Traffic Flow Management Unit (CATFMU)*
- Air Force Movement Liaison Unit (AF-MLU)

All the partners are required to work in close collaboration to ensure the successful operation of A-CDM, Kolkata.

1.2 The efficiency of the Air Transport System is highly dependent on traffic predictability. A-CDM, Kolkata effectively enhances predictability (this reduces buffer times for resource planning and flight times), overall efficiency and punctuality by linking and sharing of accurate and timely information amongst Aircraft Operators, Airport Operator, ATC, etc.

1.3 A-CDM application is developed in-house by Airports Authority of India.

1.4 A-CDM system processes and utilizes the following information from the various sources mentioned below:

a) AFTN:

i) Flight plan data

ii) EOBT

iii) ATD

iv) ETAs

v) NOTAMs

vi) METARs

vii) ADC Numbers

b) Airport Operations Data Base (AODB) of Kolkata

i) Parking stand of Aircraft

ii) Registration marking

iii) Boarding status of departing aircraft

iv) Actual “Off Block” Times

v) Actual “In-Block” Times

c) A-CDM system database

i) Arrival and Departure Taxi timings from each parking stand

ii) Runway-in-Use and Closures

iii) Hourly handling capacity of runways

iv) Priority of flights e.g. VVIP and Emergencies etc.

v) Flight schedules

1.5 A-CDM is a Common Information Sharing Platform introduced to process timely and accurate information for quick and precise decision making. A-CDM takes into consideration the ETAs of arriving aircraft, Target Off Block Times of departures, Runway-in-use and the handling capacity of the runway and determines the arrival and departure sequence. Based on the sequence and the parking position, it calculates the Target Take-Off Times & Target Start up Times of departures and Target In-Block Times for arrivals.

1.6 Implementation of ACDM will result in considerable benefits to stakeholders. The various benefits of A-CDM includes are as follows:

- Improvement in efficiency of operations
- Reduction of delays to departures at holding point.
- Orderly flow of traffic.
- Reduction in wastage of Aviation fuel.
- Reduction in Carbon Emission, reduction in Environmental Pollution.
- Optimization of airport and airspace capacity Reduction of R/T congestion
- Enhancement of safety due to reduced R/T congestion.
- Better Situational Awareness and increased confidence amongst ACDM Partners.
- Adherence to ADC (Air Defence Clearance) by stake holders.

- Better Passenger experience.
- Optimized use of ground handling resources
- Optimized use of parking stands, gates and terminals.
- Reduced apron and taxiway congestion
- Access to current MET information.

2. TWO KEY PARAMETERS OF A-CDM, Kolkata: TOBT and TSAT

2.1 TARGET OFF-BLOCK TIME (TOBT)

2.1.1 Definition: The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up/push back immediately upon reception of clearance from the SMC (GROUND). It is the real time target of ensuring readiness of an aircraft and therefore, more accurate than the static departure time mentioned in the flight plan and hence, an ideal milestone to be used by all airport partners.

2.1.2 Purpose: Air Traffic Management (ATM) based on only FPL-data is insufficient to guarantee smooth and efficient operations on the ground and in the air. The main reason for this is the incorrect updating/no updating at all, of the Estimated Off Block Time (EOBT) of the aircraft. The result is that airport partners will have an incorrect idea about the actual status of that specific aircraft. The purpose of Target Off-Block Time (TOBT) is to provide a timely, accurate and reliable estimate of an aircraft's off block time to the A-CDM, Kolkata partners. Accurate TOBTs enhance operations on the ground as they provide all airport partners with a clear picture of the intentions of aircraft on the ground.

2.1.3 Communication of TOBT: Aircraft Operators shall revise their EOBTs via AFTN using CHG/ DLA message or inform ARO for revised EOBT, who in turn will originate CHG / DLA message for transmission through AFTN.

2.1.3.1 In case No delay is expected: If there is no revision of EOBT, EOBT will be treated as the TOBT. For such cases there is no need of any communication from the Aircraft Operator to the ARO.

2.1.3.2 In case delay is expected:

a) When Air Defence Clearance (ADC) is valid: If delay is within the validity period of ADC, EOBT should be revised through CHG/DLA message. Optionally TOBT can be changed in ACDM interface. But if due to some reason coordination is not possible by both the above methods then EOBT can be revised by informing ARO/MCD.

b) When Air Defence Clearance ADC is NOT valid:

i) If delay is outside the validity period of ADC, EOBT shall be revised through CHG/DLA message or after having coordinated with MLU and obtaining the revised ADC number, by informing ARO/MCD about revised EOBT and ADC. (Option of change of TOBT will not be available in this case).

ii) ADC Validity Period: (-) 15 minutes to (+) 45 minutes of EOBT in RPL/FPL or subsequent revision of EOBT by DLA/CHG message.

Note: Once Target Start-up Approval Time (TSAT) is allocated by ATC at TOBT - 30 minutes, the TOBT can be updated thrice.

2.1.4 Accuracy: TOBTs must be updated to an accuracy of +/- 5 minutes.

2.2 TARGET START UP APPROVAL TIME (TSAT)

2.2.1 Definition: The time is calculated by A-CDM application taking into account TOBT, *CTOT** and/or the traffic situation that an aircraft can expect to start-up/push back for departure.

Note: In case there is no congestion, ATC may allow the aircraft to start, where Actual Start-up Approval Time (ASAT) will be before the TSAT.

2.2.2 Purpose: TSATs provide an optimized start-up sequence, leading to Pre-Departure sequence. TSATs reduce queuing times at the runway holding points, while ensuring optimised utilisation of the runway. ATC will continue to optimize the departure order for creating the right mix of traffic.

2.2.3 Calculation of TSAT: TSAT is calculated automatically by A-CDM Application. The TSAT is calculated by taking into account TOBT, priority of aircraft, variable taxi times, and the runway capacity. Any revision of EOBT will trigger the A-CDM application to recalculate a new TSAT for that flight as well as to re-allocate earlier TSAT to next flight which meets the requisite conditions. TSAT is displayed in various ACDM interfaces in the following colour coded information.
Blue colour: TSAT -15 minutes up to TSAT -5 minutes' Green colour: TSAT -5 minutes up to TSAT +5 minutes Red colour: After TSAT +5 minutes.

2.2.4 Generation of TSAT: TSAT is generated by ACDM interface at TOBT - 30 minutes.

2.2.5 Communication of TSAT: TSAT will be displayed in various A-CDM interfaces. The AO/GHA shall advise flight crew of TSAT, displayed in the A-CDM interface.

2.2.6 Priorities for generation of TSAT:

The following aircraft will be accorded priority by A-CDM application at the time of generating TSATs:

a) VVIP flights
b) Aircraft in emergency
c) Ambulance aircraft
d) Aircraft in relief missions
e) Aircraft returning to stand after push back/taxiing out for any reason.

2.2.7 Airlines/GHAs are required to monitor the A-CDM interface regularly to get information on the revised TSAT if any, in respect of their aircraft. It may please be noted that the earlier allocated TSAT may get changed due to the following reasons:

i) Change of runway
ii) Change in taxi times
iii) Revised TOBT of the subject aircraft
iv) Cancellation/ revision in TOBT of other aircraft in the departure sequence.
v) Priority handling
vi) Unusual Occurrences, etc.
vii) Change of aircraft type*
viii) Application of ATFM regulation or ATC restriction resulting in new CTOT or start up delay*
ix) Inclement Weather*

x) Bird activity*

xi) Change of route (SID)*

2.2.8 Accuracy: The TSAT will remain valid for +/- 5 minutes.

3 Co-ordination with the Central Air Traffic Flow Management Unit (CATFMU)*

3.1 A permanent and fully automatic data exchange with the ATFMU will be established. The data transfer will enable highly accurate early predictions of landing and departure times. *

3.2 *Furthermore, it allows for more accurate and efficient CTOT calculations due to the use of more accurate local target take-off times.* *

4 TOBT and TSAT based A-CDM, Kolkata Start-up/Push-back procedures

4.1 Pre-Departure Sequence: Pre-Departure sequence is the order, in which the ATC plans the aircraft to depart from their Gate/Stands. It should not be confused with the Pre-Take-off sequence where ATC decides on the order in which the aircraft at holding points of the runway will depart.

4.2 Start-Up and Push-Back Procedure:

i. The pilot will contact Clearance Delivery (CLD) to request en-route clearance and SID between TSAT - 15 minutes to TSAT - 5 minutes. Blue Zone.

ii. The aircraft must be ready to Start-up/Push-back at TOBT and request SMC (GROUND) for Start-up/Push-back at TSAT – 5 minutes and TSAT + 5 minutes. Green Zone.
(Note: The system may prepone TSAT up to TOBT)

iii. In case for Schedule Flight, “Boarding” has not started at or before TSAT – 5 Minutes, allotted TSAT will be cancelled and associated slot will be allocated to next eligible aircraft.

a. A new TOBT is required from Airlines Operator.

b. On receipt of new TOBT, the flight will be re-sequenced according to new TOBT and a new TSAT with a subsequent delay will be issued.

iv. If at TSAT +5 minutes: Red Zone, ATC has not received Start-up/Pushback request, the aircraft will lose its position in sequence.

a. A new TOBT is required from Airlines Operator.

b. On receipt of new TOBT, the flight will be re-sequenced according to new TOBT and a new TSAT with a subsequent delay will be issued.

v. ATC should normally be able to issue start-up/push-back at TSAT. Pilots will be informed by ATC of any revised TSAT if there is a delay to TSAT in excess of 5 minutes.

vi. Taxi clearance must be requested within 5 minutes of Start-up/Pushback approval time.
(Note: If this has not occurred, SMC (GROUND) must be notified of the extent of delay. In such cases, aircraft may lose its departure slot and a new TOBT may be required in A-CDM, Kolkata interface. AO shall be responsible to obtain new ADC number from IAF-MLU).

vii. The Pre-Departure (Off- block) Sequence will be determined in accordance with Target Start up Approval Time (TSAT) and NOT in accordance with the Start-up Request. Pre-Departure Sequence will not have any bearing on Actual Departure sequence.

(Note: Actual Departure sequence may differ from pre-departure sequence in order to optimize the Runway and Airspace utilization.)

** [Future provision kept for incorporating with India – Central Air Traffic Flow Management]*

5 TERMS AND ABBREVIATIONS

All Acronyms are time parameters which have a standard length of four characters and have been arranged as per aircraft movement sequence.

[While developing the A-CDM, Kolkata procedures, the guidance material published by Euro control, ICAO A-CDM and other Organizations has been used].

S. NO.	ACRONYMS	DEFINITION	EXPLANATION
5.1	ELDT	Estimated Landing Time	The estimated time that an aircraft will touchdown on the runway. (Equivalent to ATC ETA = Estimated Time of Arrival = landing).
5.2	ALDT	Actual Landing Time	The time that an aircraft lands on a runway. (Equivalent to ATC ATA = Actual Time of Arrival = Landing, ACARS = ON).
5.3	EXIT	Estimated Taxi- In Time	The estimated taxi time between landing and inBlock.
5.4	AXIT	Actual Taxi-In Time	AXIT = AIBT – ALDT
5.5	SIBT	Scheduled InBlock Time	The time that an aircraft is scheduled to arrive at its first parking position.
5.6	EIBT	Estimated InBlock Time	The estimated time that an aircraft will arrive inblocks. (Equivalent to Aircraft Operator's ETA = Estimated Time of Arrival).
5.7	AIBT	Actual In-Block Time	The time that an aircraft arrives in-blocks. (Equivalent to Aircraft Operator's ATA = Actual Time of Arrival, ACARS = IN).
5.8	STTT	Scheduled Turnaround Time	STTT = SOBT – SIBT
5.9	ETTT	Estimated Turnaround Time	The time estimated by the AO/GH on the day of operation to turnaround a flight taking into account the operational constraints.
5.10	MTTT	Minimum Turnaround Time	The minimum turnaround time agreed with an AO/GH for a specified flight or aircraft type.
5.11	ATTT	Actual Turnaround Time	ATTT = AOBT – AIBT
5.12	ASRT	Actual Start Up Request Time	Time the pilot requests start up clearance.
5.13	TSAT	Target Start Up Approval Time	The time provided by ATC taking into account TOBT, CTOT* and/or the traffic situation that an Aircraft can expect start-up / push back approval. Note: The actual start up approval (ASAT) can be given in advance of TSAT.
5.14	ASAT	Actual Start Up Approval Time	Time that an aircraft receives its start-up approval.
5.15	SOBT	Scheduled OffBlock Time	The time that an aircraft is scheduled to depart from its parking position.
5.16	EOBT	Estimated OffBlock Time	The estimated time at which the aircraft will start movement associated with departure (ICAO)

5.17	TOBT	Target OffBlock Time	The time that an Aircraft Operator estimates that an aircraft will be ready, all doors closed, boarding bridge removed, push back vehicle available and ready to start up / push back immediately upon reception of clearance from the SMC (GROUND).
5.18	AOBT	Actual OffBlock Time	Time the aircraft pushes back / vacates the parking position. (Equivalent to Aircraft Operator's ATD = Actual Time of Departure & ACARS = OUT)
5.19	ARDT	Actual Ready Time (for Movement)	When the aircraft is ready for start-up/push back or taxi immediately after clearance delivery, meeting the requirements set by the TOBT definition.
5.20	EXOT	Estimated TaxiOut Time	The estimated taxi time between off-block and take off. This estimate includes any delay buffer time at the holding point prior to take off.
5.21	AXOT	Actual Taxi-Out Time	AXOT = ATOT – AOBT
5.22	ETOT	Estimated Take Off Time	The estimated takeoff time taking into account the EOBT plus EXOT.
5.23	CTOT*	Calculated Take Off Time*	A time calculated and issued by the appropriate Central Management unit, as a result of tactical slot allocation, at which a flight is expected to become airborne. (ICAO Doc 7030/4 – EUR, Table 7)*
5.24	TTOT	Target Take Off Time	The Target Take Off Time taking into account the TOBT/TSAT plus the EXOT. Each TTOT on one runway is separated from other TTOT or TLDT to represent vortex and/or SID separation between Aircraft.
5.25	ATOT	Actual Take Off Time	The time that an aircraft takes off from the runway. (Equivalent to ATC ATD = Actual Time of Departure, ACARS = OFF).

II. STANDARD AIRPORT TAXI ROUTES (SATR)

1. INTRODUCTION:

Standard Airport Taxi Routes are being designed to enhance ground control operations by using standard taxi instructions for both arriving and departing traffic. SATR is recommended as one of the 'Air Traffic Control best practices', in 'The manual on the prevention of Runway Incursions', (DOC 9870 Appendix-C). A single taxi instruction comprising of a number of taxiways may lead to read back errors, occurrence and correction of which increases R/T congestion and affects ground control operations as well as traffic handling capacity of ATCO.

2. PURPOSE:

The purpose of Standard Airport Taxi Routes (SATR) is to use standard taxi routes instead of long and ambiguous taxi instructions. SATR provides continuous taxi instructions including smooth transitioning of frequency from one surface movement controller to another. This alleviates R/T congestion and will thus allow controller to have better surveillance of the traffic in peak traffic times. Theimplementation of SATR will lead to reduction in workload of both ATCOs and Pilots.

3. CONCEPT:

3.1. A single SATR comprises of a number of taxiways which together form complete taxi instructions to a departure/arrival.
3.2. SATR have been designed for RWY 19L, RWY 19R, RWY 01R, RWY 01L.
3.3. ARRIVAL SATR: The arrival route starts after the exit taxiway of Runway and terminates prior to a hotspot or Taxiway-Apron taxi-lane junction, for which the route is defined. The arrival follows the SATR and holds short of hotspot or Taxiway-Apron taxi-lane junction for its parking bay. Further taxi will be advised by the SMC as per the designated parking stand.
3.4. DEPARTURE SATR: The departure route starts from the Apron taxi-lane for which the route is defined and terminates at the designated holding point of the specific Runway or Intermediate Holding Position. The departure will join the route via the apron lane as advised by the ATC.
3.5. Departures/Arrivals will join/leave SATR via Apron lanes as instructed.

4. NOMENCLATURE:

Route [NN] [A/D] [N]

- NN: Two-digit Runway Designator
- A/D: Arrival/Departure
- N: Different Taxi Route

Examples:

- Route19D2: Departure route for RWY 19L starting from aircraft parking stand to Holding Point RWY 19L.
- Route01A3: Arrival route of RWY01R terminating on RWY 19R (Short of TWY E).).

5. SATR

A total of 14 SATR are designed:

5.1 SATR for Arrivals (09)

S. No.	ARR SATR NAME	RUNWAY	ROUTE DETAILS	REMARKS
1.	Route19A1	19L	VACATE VIA TWY R1 TURN LEFT-TAXI VIA RWY 19R. HOLD SHORT OF TWY E ON RWY 19R	EXPECT FURTHER TAXI VIA E OR RWY 19R.
2.	Route19A2	19L	VACATE VIA TWY R CROSS RWY 19R-TWY C, HOLD SHORT OF TWY A	EXPECT FURTHER TAXI VIA A. (*Left or Right Turn*) OR G
3.	Route19A3	19L	VACATE VIA TWY R. TURN LEFT-TAXI VIA RWY 19R. HOLD SHORT OF TWY H ON RWY 19R.	EXPECT FURTHER TAXI VIA H OR J.
4.	Route19A4	19R	VACATE VIA TWY H. TAXI VIA H1. HOLD SHORT OF INTERSECTION OF TWY F1 & F2	EXPECT FURTHER TAXI VIA F1 OR F2
5.	Route01A1	01R	VACATE VIA TWY Q1 TURN RIGHT, TAXI VIA RWY 01L-TWY N. HOLD SHORT OF INTERSECTION OF TWY N1, N2, N3 AND N4 ON TWY N.	EXPECT FURTHER TAXI VIA N1 OR N2 OR N3 OR N4
6.	Route01A2	01R	VACATE VIA TWY Q2 CROSS RWY 01L-TAXI VIA TWY N. HOLD SHORT OF INTERSECTION OF TWY N1, N2, N3 AND N4 ON TWY N	EXPECT FURTHER TAXI VIA N1 OR N2 OR N3 OR N4
7.	Route01A3	01R	VACATE VIA TWY A TURN LEFT-TAXI VIA RWY19R. HOLD SHORT OF TWY E ON RWY19R.	EXPECT FURTHER TAXI VIA E OR RWY 19R.
8.	Route01A4	01L	VACATE VIA TWY N. HOLD SHORT OF INTERSECTION OF TWY N1, N2, N3 AND N4 ON TWY N.	EXPECT FURTHER TAXI VIA N1 OR N2 OR N3 OR N4.
9.	Route01A5	01L	TURN LEFT TO VACATE VIA TWY A. TAXI VIA TWY A1- TURN LEFT TO JOIN TWY N4. HOLD SHORT OF TWY F.	EXPECT FURTHER TAXI VIA F. ALL ACFT UPTO CODE C ONLY.
Note: 1. Arriving aircraft vacating runway shall contact ground on frequency 121.9 MHZ or 121.725 MHZ unless instructed otherwise by ATC Tower. 2. Aircraft to vacate runway via exit taxiway as assigned by tower or inform inability to ATC by 10 NM to touchdown.				

5.2 SATR for Departures (05)

S. No.	DEP SATR NAME	RUNWAY	ROUTE DETAILS	REMARKS
1.	Route19D1	19L	TAXI VIA H1-TURN LEFT ON TWY A. TAXI VIA TWY N4-TWY N-TWY P. HOLD AT HOLDING POINT RWY 19L ON TWY P.	JOIN ROUTE VIA F2/F1.
2.	Route19D2	19L	TAXI VIA TWY N2-TWY N-TWY P. HOLD AT HOLDING POINT RWY 19L ON TWY P.	JOIN ROUTE VIA F2/F1/F.
3.	Route19D3	19R	TAXI VIA TWY N2-TWY N. HOLD AT HOLDING POINT RWY 19R ON TWY N	JOIN ROUTE VIA F2/F1/F ACFT UPTO CODE C ONLY..

4.	Route01D1	01R	TAXI VIA TWY H1-TWY H-TURN LEFT ON RWY 01L-TURN RIGHT ON TWY D. HOLD AT HOLDING POINT RWY 01R ON TWY D.	JOIN ROUTE VIA F2/F1.
5.	Route01D2	01L	TAXI VIA J1, TURN RIGHT ON TWY A. HOLD AT HOLDING POINT RWY 01L ON TWY A	JOIN ROUTE VIA F2/F1.

Note:

1. Aircraft, instructed to monitor Kolkata Tower (118.100 MHz), will not transmit on 118.1 MHZ unless being called by Tower or when required for safety of aircraft.
2. After two-way communication is established with Kolkata Tower, the aircraft thereafter shall be deemed to be changed over to Kolkata Tower.

6. OPERATING PRINCIPLE:

6.1 SATR shall be used to describe taxi routes.

6.2 All aircraft shall have positive clearance before crossing/entering a runway.

6.3 During 'Low Visibility Procedure' aircraft shall follow taxi routings as mentioned in sub-para 4.6 of para II. LOW VISIBILITY PROCEDURES in VECC AD 2.23 of eAIP India or as instructed by ATC.

6.4 Change of holding point for a departure (in case of request or traffic requirement), other than that issued in SATR shall be made well in advance and revised instructions should be given for the same.

6.5 ATCOs may use a combination of SATR along with progressive taxi instructions on a tactical basis to maintain an efficient flow of surface movements.

6.6 In case of traffic or closure of a taxiway or a portion thereof, the controller has the flexibility to cancel the use of SATR at any time and use progressive taxi instructions.

6.7 If for any reason, aircraft crew is unable to comply or understand the SATR instructions; timely request should be made for progressive taxi instructions.

6.8 In case ATCO cancels SATR for any aircraft, R/T instruction - Cancel (SATR Name) Taxi via (revised taxi route) should be used.

6.9 An aircraft may be advised to join a SATR via an intermediate taxiway. Such instruction will be given by defining the taxiway as a prefix to SATR.

Example: Taxi via D-Route19A3 means VACATE VIA D-TURN LEFT-TAXI VIA RWY 19R. HOLD SHORT OF H JUNCTION.

7. PHRASEOLOGY:

7.1 **For Departures**

(Aircraft Call sign) Kolkata Ground: Taxi to holding point Runway [Designator] via [taxiway joining respective parking bay to SATR] [SATR] Time [time]

e.g. AIC (XYZ) KOLKATA GROUND: TAXI TO HOLDING POINT RWY19L VIA ROUTE19D1TIME 0200Z.

7.2 For Arrivals

(Aircraft Call sign) Kolkata Ground: Taxi to bay [number] via [SATR] exit [taxiway joining respective SATR to parking bay]

e.g. AIC (XYZ) KOLKATA GROUND: TAXI TO BAY 52 VIA ROUTE19A1 EXIT H-H1-F1.

7.3 **Revised Taxi Instruction after issuing SATR.**

(Aircraft Call Sign) KOLKATA GROUND: CANCEL SATR, TAXI VIA... (REVISED ROUTING).

III. The procedure for reducing RT congestion.

1. Departure aircraft requiring back-track on runway shall inform ATC prior to taxi requirement.
2. Deperature aircraft passing 1000 Feet to change over to kolkata radar (PRI FREQ 127.9MHZ) , unless otherwise specified by ATC Tower.
3. Departing aircraft, on first contact with approach, shall transmit only the call sign, alterante passing and the assigned standard instrument departure.
4. An aircraft being changed over from other units of Kolkata Radar shall transmit only its call sign in first contact with approach.
5. Aircraft, instructed to monitor tower frequency, will not transmit unless being called upon by tower or when necessary considering aircraft safety. it is deemed that aircraft shall change over to tower thereafter.
6. Arrival aircraft vacating runway to contact ground on frequency 121.9MHZ unless instructed by ATC tower.
7. Aircraft to vacate runway via exit taxiway as assigned by tower on final approach or inform inability to atc.
8. Aircraft shall comply with all speed restrictions unless removed by atc.
9. Arrival and departure aircraft to minimize runway occupancy time for enhancing runway capacity.
10. Aircraft shall be ready for immediate takeoff roll upon receipt of takeoff clearance when line up and takeoff clearance is issued simultaneously aircraft shall do so in a constant movement.

IV. No Aircraft shall take lock turn/ pivotal turn on the Runway to avoid pavement damage.

V. PUSHBACK AND STARTUP PROCEDURE ON APRON-3

1. The objective of the pushback and start-up procedure is to reduce ground conflicts in pushback and start-up of aircraft, reduce ground delays and reduce radiotelephony congestion thereby enhancing the operational efficiency of Kolkata Airport.

2. These procedures shall apply to activities of all agencies involved in aircraft pushback & start-up on Apron-3 and will be applicable to Code-C and below type of aircraft only. Aircraft, pushing back, facing east on TWY-F2, from parking stand 58R to 61L and 33 to 40, are divided, zone wise, in three groups (Gr. 1, Gr. 2 and Gr. 3) as mentioned in the pushback procedure table.

3. Kolkata Ground (frequency 121.9/121.75 MHz, or any alternate frequency) shall be contacted for pushback and start-up permission only after clearance is received from Clearance Delivery and the Pilot-In-Command has established that the aircraft is ‘ready’ for pushback/start-up.

Note: For the purpose of clarity, ‘ready’ means all doors of the aircraft are closed, the Passenger Boarding Bridge (s) (PBB) / Step Ladder (s) are disconnected / withdrawn, the tug is connected to the aircraft and the ground engineer is in position and in contact with the Pilot-In-Command (PIC).

4. On receiving the above-mentioned information, ATC will allow pushback and start-up of the aircraft depending on the traffic.

5. Tug release points have been provided with nomenclature for tug release pints beginning with letter “T” followed by a numeral. The location of Tug release point T1 is abeam stand 60L, T2 is abeam stand 59 and T3 is abeam stand 58R.

6 (a). A gap of at least two stands has to be ensured between two Code C and below type of departures pushing back simultaneously. For example, if aircraft pushes back from 58L, then pushback won’t be possible from 59R and 59L and next possible pushback will be from 60R. Such simultaneous Pushback for Tug release points T1, T2 and T3 can be approved for one side of stands only, either north (58R to 61L) or south (33 to 40) of F2. For pushback from either side of F2, i.e., north and south of F2, simultaneous pushback for Tug release points T1 and T3 is only allowed. Pushback will

normally be given based on the least average delay principle. Thus, pushback for T3 will not normally be given after pushback for T1 or T2 and pushback for T2 will not normally be given after pushback for T1, unless they are asked simultaneously.

6(b). For code D or higher type of aircrafts, for Pushback from stand 58, Tug release point T3, from stand 59 Tug release point T2 and from stand 60 & 61, Tug release point T1 shall be used to release tug. For code D or higher type of aircrafts, simultaneous push back can be approved for Gr. 1 and Gr. 3 aircrafts for Tug release point T1 and T3.

7. Arriving Code “C” and below type of aircrafts can be allowed to park to Stands of Gr. 2 & Gr. 3, if there is a pushback from stands of Gr. 1. Similarly, arriving Code C and below type of aircrafts can be allowed to park in to the stands of Gr 3, if there is a pushback from stands of Gr. 2 for Tug release Point T2. If there is a push back of code D or E type of aircraft from Gr. 1 for Tug release point T1, arriving Code D or E type of aircraft can be allowed to park in to the stand of Gr. 3.

8. Simultaneous pushback and start-up can be approved for a single code C and below type of aircraft.

9. In case of aircraft parked in stand 57L and pushback facing east in F2, then it should be dealt with the procedures of stands in Gr.3

10. Aircraft can dock into 57/57L, even if there is a departure at tug release point T3 for all categories.

11. Pushback of 57/57L can be permitted in F1/J1, even if there is a pushback for the tug release point T3 for all categories.

12. If aircraft commenced pushback from stands of Gr. 3 (up to Tug release point T3), second pushback from stands 57 and 57L on F2 is not possible.

13. The pilot shall adhere strictly to SOP by aircraft manufacturer/ relevant operational manuals for starting up engine(s) at idle power or breakaway at locations as described below.

14. At all times ground crew must strictly follow the safety instructions on hazard zones set out by aircraft manufacturer / relevant operational manuals.

15. The vehicular movement behind parking stand 61L shall be stopped by the concerned ground crew till the aircraft from the stand 60, 60L, 61R, 61, 61L 38, 39 & 40 are at Tug release point T1.

16. Deployment of wing walkers/wing guards by airline is mandatory prior to undertaking pushback from any stand.

17. On receipt of the clearance, the PIC shall read back the pushback clearance given by ATC, then co-ordinate with the ground crew for the pushback and start-up of the aircraft. The ground crew must ensure that the area in the front, behind and around the aircraft is clear of personnel, vehicles, equipment and other obstructions before commencing pushback and start-up of the aircraft.

18. The PIC may start one engine, on idle power, before commencing pushback on the aircraft stand, in coordination with the ground crew. Starting of 2nd engine on idle power, before commencing pushback on the aircraft stand, shall be permitted after PIC confirms to ATC that the 2nd engine is being started with appropriate power unit and necessary ground precautions have been ensured. Aircraft’s nose wheel to stop on relevant Tug release point.

19. No cross-bleed start up by aircraft is permitted till the pushback and/or pull ahead procedure is complete, and the Aircraft is aligned with the taxilane / taxiway centreline marking. Prior to starting the cross bleed, PIC shall seek confirmation from the ground crew for hazard free zone. At all times ground crew must follow the safety instructions on hazard zones set out by aircraft manufacturer/ relevant Operational manuals.

20. Ground crew of aircraft intending to do a cross bleed start and must assess the clearance behind & in-front of aircraft and in case they feel that there is insufficient clearance, they shall request SMC for repositioning the aircraft to a position from where cross bleed start can take place. As a broad guideline, a separation distance equivalent to 3 (three) aircraft length should be clear behind the aircraft so that there is no jet blast impact behind the aircraft due to cross bleed start.

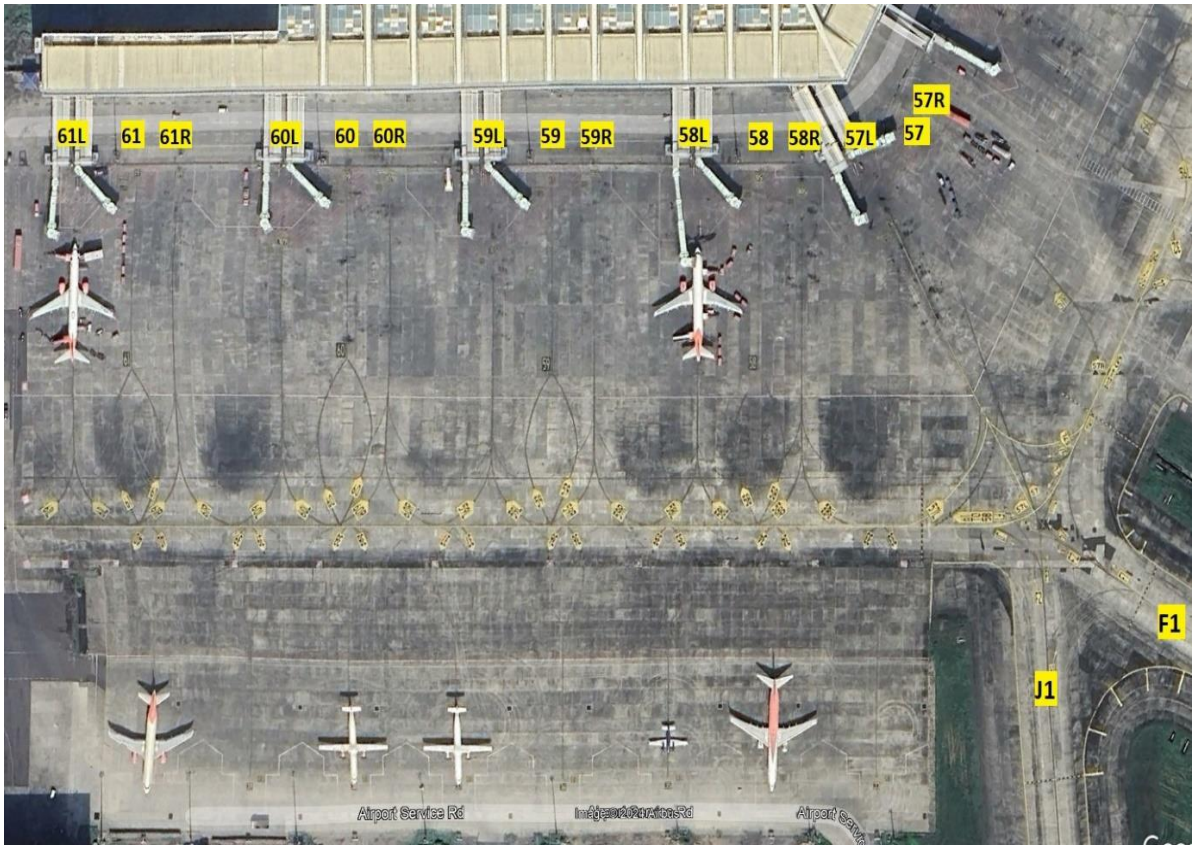
21. Vehicle or movement of personnel is NOT permitted behind the aircraft once the anti-collision beacons of the aircraft

have been switched ON.

22. Pilots shall adhere to the pushback and start up procedures and will use minimum breakaway power.

PUSHBACK PROCEDURES

Grouping	P/S	Pushback Procedure	Remarks
Gr. 3	58R, 58 & 58L	Push back on TWY-F2 & pull ahead up to Tug Release Point T3.	Only one aircraft to push back from stand 58R, 58, 33, 34 & 35 58L, 33 to 35
	33, 34 & 35		
Gr. 2	59R, 59, 59L & 60R	Push back on TWY-F2 & pull ahead up to Tug Release Point T2	Only one aircraft to push back from stand 59R, ,59, 36 & 37 59L, 60R, 36 and 37
	36 & 37		
Gr. 1	60L, 61R, 61L,60,61	Push back on TWY-F2 & pull ahead up to Tug Release Point T1.	Only one aircraft to push back from stand 60L, 60,61R, 61 61L, 38, 39 and 40
	38, 39 & 40		



VI UTILIZATION OF PARALLEL TAXIWAYS N1, N2, N3 AND N4

i) NSCBI Airport has four parallel taxiways (TWYs N1, N2, N3 & N4) towards western side of TWY A. These four parallel TWYs merge on the northern side into a single TWY N, which connects the beginning of RWY 19R and TWY P. TWY P connects to RWY 19L.

ii) TWY N2 & TWY N4 are CAT III compliant.

RWY in use	Type of aircraft	Taxiway in used	No. of Taxiway in simultaneous use	Remarks
Primary RWY (19L-01R)	All code C	N1, N3 and N4	03	N2 cannot be used
	Code E and Code C	N2: Code E N4: Code C Or N2: Code C N4: Code E	02	N1 and N3 cannot be used
	Code E and Code C	N1: Code C N4: Code E	02	N2 and N3 cannot be used
Secondary RWY (19R-01L)	All code C	N1, N3 and N4	03	N2 cannot be used
	Code E and Code C	N2: Code E N4: Code C	02	N1 and N3 cannot be used.

Pushback and Taxiing Procedures:

1. Direction of push back from parking stands 101 to 113 and 114/115 (L, R) facing north or facing south to be given only on TWY N1. (TWY N1 is for taxing of aircraft up to code C)
2. Direction of push back from parking stands 114 and 115 facing north or south shall be given only on TWY N2 (TWY N2 is for taxing of aircraft up to code E)
3. No simultaneous pushback shall be given from adjacent parking stands.
4. Simultaneous pushback is permitted only from alternate parking stands with caution.
5. When pushback is given on TWY N1, there is no restrictions for taxing of aircraft on TWY-N3 & N4. However, no taxing of aircraft shall be given on TWY N2.
6. When pushback is given from parking stands 114 and 115 on TWY N2 aircraft movement up to Code C is permitted to taxi on TWY-N4. No taxing of aircraft shall be given on TWY N1 and TWY N3.
7. Idle power engine run up on parking stands by airline operators should be carried out in coordination with AOCC for issuance of necessary instruction by Tower.
8. For RWY-19R, Code-D & E type of aircraft shall hold at intermediate holding position on TWY-N abeam 19R glide path.
9. Simultaneous operations on portion of Taxiway A (between TWY F and RWY 19 R holding point) and TWY N4 shall not be permitted.
10. When the runway in use is RWY 19R, aircraft requesting departure from TWY A intersection shall inform the same to ATC prior to commencement of taxi.

Docking Procedure for arrival aircraft:

1. Up to Code-C type of aircraft which will be docking to parking stand 101 to 113 and 114/115(L, R) should use TWY-N1.
2. For Code-D/E type of aircraft which will be docking to parking stand 114 and 115 should use TWY-N2.

Link TWY-A1, B1 and B2 are intersecting perpendicularly to TWY-N1, N2, N3 & N4. These shall be used to connect parallel taxiways (TWY- N1, N2, N3 & N4) from TWY-A and TWY-B

VECC AD 2.21 NOISE ABATEMENT PROCEDURES

Consistent with safety of aircraft operations and in consideration of high intensity runway operations, pilots should minimize the use of reverse thrust after landing to reduce disturbance in areas adjacent to the aerodrome.

VECC AD 2.22 FLIGHT PROCEDURES

I. Kolkata TMA Routing

Route Designator	Kolkata TMA Routing
P646	DOPID N205503.2 E 0891216.1 – 339 DEG/112NM ---CEA VOR N223842.6 E0882710.4
M773	BUBKO N191103.7 E0883950.5 – 349 DEG/151NM -- LEGOS N213802.9 E0880520.6 – 019 DEG/064NM --- CEA VOR N223842.6 E0882710.4

II. RADIO COMMUNICATION FAILURE PROCEDURES:

1. **INTRODUCTION:**

The Radio Communication Failure procedures established herein are based on the provisions of ICAO PANS ATM DOC 4444. The objective is to standardize the actions to be taken by the pilot of arriving and departing aircraft experiencing radio communication failure at Kolkata airport, when ATS surveillance system is used in the provision of air traffic control in such airspace.
During the periods of non-availability of ATS surveillance system in part or whole of the concerned airspace in Kolkata, the provisions of ICAO PANS ATM DOC 4444 shall be followed.

2.**SCOPE:**

The procedures in this document shall be applicable to the pilots of arriving and departing aircraft experiencing radio communication failure at Kolkata airport when ATS surveillance system is used in the provision of air traffic control in such airspace.

3.**GENERAL:**

3.1 All Transponder equipped aircraft experiencing Radio Communication Failure shall set transponder to Mode A/C code 7600 as soon as practicable. (Note: This Requirement of Setting transponder to Mode A/C code 7600 in no way imposes any restriction on the pilot's decision to set transponder to Mode A/C code 7500 or 7700, whenever required).

3.2 An aircraft equipped with other surveillance system transmitters, including ADS-B and ADS-C, might indicate the loss of air-ground communication by all of the available means.

3.3 Aircraft equipped with CPDLC should log on to Kolkata Data-link (ADS-CPDLC) VECF, so that CPDLC connection may be established for communication.

3.4 To indicate that it is experiencing Radio Communication Failure, an aircraft equipped with ADS-C shall transmit appropriate message in this regard.

3.5 Immediately after detection of RCF, the pilot shall try to communicate to the concerned ATS unit through the frequencies of adjacent units/stations, or 121.5 MHz or by relay through other aircraft. The RCF aircraft shall also try to contact the appropriate HF frequencies to relay messages to the concerned ATS unit.

3.6 Aircraft shall avoid all Restricted Areas and active Danger Areas. Aircraft shall not enter Bangladesh airspace if flight plan does not include Dhaka FIR unless emergency conditions warrant so. If aircraft is observed entering any of the areas mentioned above concerned ATC unit shall inform the units concerned accordingly.

3.7 Pilot shall make blind transmission to ATC of all necessary reports and actions taken by the aircraft, e.g. descent, turn, proceeding to waypoint etc., irrespective of whether partial/ complete RCF has been established or not.

4. ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT:

4.1 In case of an arriving aircraft experiencing RCF, when Runway for landing has already been advised to the aircraft by ATC, such runway shall be considered as the assigned runway. If an arriving aircraft has not been advised of any runway, Runway 19L shall be considered as assigned runway for such arrival. During the notified periods of maintenance/closure of Runway 19L, Runway 19R shall be considered as assigned Runway and vice-versa.

4.2 In case of departures from Kolkata returning due to RCF, the departure runway of such aircraft shall be considered as assigned runway for landing.

4.3 Runway lights and Approach lights in 'SWITCHED ON' position shall indicate the availability of such Runway for aircraft experiencing RCF.

4.4 Irrespective of visibility/weather conditions, Runway and Approach lights in 'SWITCHED OFF' position shall indicate non-availability of such Runway for aircraft experiencing RCF.
In such cases, alternate runway shall be made available for such aircraft. Runway 19R shall be alternate to Runway 19L and Runway 01L shall be alternate to Runway 01R. Similarly, Runway 19L shall be alternate to Runway 19R and Runway 01R shall be alternate to Runway 01L.

4.5 In case strong tail wind conditions are encountered during approach for the assigned Runway, the aircraft experiencing RCF will carry out a missed approach on assigned runway and after following complete missed approach procedure for such approach, aircraft will carry out the published ILS/VOR approach procedure for the runway which is in opposite direction to the assigned Runway, e.g. Runway 01R in case of assigned Runway 19L.

5. PROCEDURE FOR ARRIVALS:

5.1 General:

a.Except when descent clearance has already been received from ATC and acknowledged, pilot shall not commence descent before 100 NM from CEA VOR or until three minutes after setting Mode A/C code 7600 whichever is later.
b.In case radio communication failure takes place after establishing ILS/ final approach track, aircraft may continue the approach and land if visual, or go around and carry out the missed approach procedure and join the VOR (112.5) CEA holding procedure climbing to and maintaining 2000 FT.

5.2 Arriving Aircraft – STAR Assigned

When STAR has been assigned to an arrival experiencing radio communication failure, it shall:
a.Continue on the assigned STAR following all level and speed restrictions applicable to STAR, as far as practicable.
b.Commence descent as in Para 5.1(a) above. Descend to FL70.
c.At the end of the STAR, descend to 2000ft and take a suitable turn to intercept localizer or final approach track of the published procedure for the assigned runway, remaining within 25 NM of CEA.
d.If additional track miles are required to adjust descent, hold across the final approach track of the assigned runway between 15 and 25 NM from CEA.

5.3 Arriving Aircraft – STAR Not Assigned

When STAR has not been assigned to an arrival experiencing radio communication failure, it shall:
a.Continue on ATS route, (re-join ATS route if given heading or flying offset)
b.Commence descent as in Para 5.1 (a) above. Descend to FL70 and proceed to CEA VOR.
c.Descend in CEA hold to 2000ft and carry out the published ILS/VOR approach procedure for the assigned runway.

5.4 Arriving Aircraft Being Radar Vectored

An aircraft which is radar vectored, on experiencing radio communication failure, shall:

a.Maintain the last assigned speed, level and heading for 3 minutes after selecting MODE A/C 7600.
b.Then proceed direct to CEA VOR descending to FL70 according to Para 5.1 (a) above.
c.Descend in CEA hold to 2000 FT and carry out the published ILS/VOR approach procedure for the assigned runway.

6. PROCEDURE FOR DEPARTURES:

6.1Departure Intending to Continue to Destination

A departing aircraft experiencing radio communication failure and intending to continue to its filed plan destination shall:

a.Continue on the assigned SID or heading, climbing to/ maintaining cleared and acknowledged level or FL60, whichever is higher.
b.Three minutes after setting Mode A/C code 7600 or reaching FL60 or cleared Flight level (if higher than FL60) whichever is later:

i.if following SID, continue on SID to join ATS route and climb to filed Flight level and continue as per the filed flight plan.
ii.if following Radar heading or have been directed by ATC to proceed offset using RNAV without a specified limit, proceed in the most direct manner possible to re-join the current flight plan route no later than the next significant point, maintaining cleared flight level/FL60 whichever is higher. After joining ATS route, climb to filed Flight level and continue as per the filed flight plan.

6.2 Departure Intending to Land Back at Kolkata:

A departing aircraft, experiencing radio communication failure and intending to land back under VMC at Kolkata Airport in accordance with provisions of DGCA CAR Series C part I, shall:
a.Continue on assigned SID or radar heading and set Mode A/C code 7600.
b.Climb/ stop climb at a level according to the table given below:

Phase of flight	Actions to be followed by pilot
At or below FL70	Maintain/Climb to FL70
Above FL70, but below FL230	Maintain/ Climb to: The last cleared and acknowledged level or FL230/ FL220, (appropriate to the direction of the flight) whichever is lower.
At or above FL230	Stop climb at the next level appropriate to the direction of the flight.

c.On reaching a suitable level according to Para 6.2 (b) above, or three minutes after setting Mode A/C code 7600, whichever is later, take a suitable turn to establish inbound the same ATS route and proceed to CEA VOR. After establishing the inbound track, descend as mentioned in Para 6.2 (d) or 6.2 (e) below.

d.At 50NM descend to FL70, if jettisoning is not necessary.

e.In case the aircraft needs to jettison fuel, it shall:

i.Climb/descend to FL100. Proceed outbound on R-170 CEA VOR until 80NM.
ii.Hold at FL100 between R-160 and R-180 CEA VOR, 80 and 100 NM from CEA DME to jettison fuel.
iii.On completion of jettisoning return to CEA VOR via R-170 CEA VOR. At 50NM inbound descend to FL70 and subsequently follow the provisions of Para 6.2 (f) below.

f.Descend in CEA hold to 2000 FT and leave CEA VOR to carry out the published ILS/VOR approach procedure for the assigned runway.

III. ADS-B based Enroute ATS Surveillance services are provided to suitably equipped aircraft on opportunity basis within the coverage area of Dibrugarh, Agartala, Guwahati, Nagpur, Varanasi, Patna, Bhubaneswar and Port Blair ADS-B sensors in the jurisdiction of Kolkata Area Control.

ADS-B equipped aircraft flying within the jurisdiction of Kolkata Area Control to comply with relevant provisions.

IV. <u>RNAV – 1 (GNSS) SIDs AND STARs:</u>

1. DETAIL OF STARS:

No.	ROUTES	STAR NAMES (CODE)			
		19L	**19R**	**01R**	**01L**
1	A791/G450/Q19	JJS 2A	JJS 2B	JJS 2G	JJS 2H
2	R460	TEPAL 2A	TEPAL 2B	TEPAL 2G	TEPAL 2H
3	W52	SEGMO 2A	SEGMO 2B	SEGMO 2G	SEGMO 2H
4	R325	DUMKA 2A	DUMKA 2B	DUMKA 2G	DUMKA 2H
5	A467	ONOTO 2A	ONOTO 2B	ONOTO 2G	ONOTO 2H
6	R472/R598	AGODA 2A	AGODA 2B	AGODA 2G	AGODA 2H
7	A462	BEMAK 2A	BEMAK 2B	BEMAK 2G	BEMAK 2H

8	B593	NOKAT 2A	NOKAT 2B	NOKAT 2G	NOKAT 2H
9	B465	SUMAG 2A	SUMAG 2B	SUMAG 2G	SUMAG 2H
10	L507	AVPOP 2A	AVPOP 2B	AVPOP 2G	AVPOP 2H
11	W112/P646	DOPID 2A	DOPID 2B	DOPID 2G	DOPID 2H
12	Q10	GOLAN 2A	GOLAN 2B	GOLAN 2G	GOLAN 2H
13	W41/A465	KAKID 2A	KAKID 2B	KAKID 2G	KAKID 2H
14	M773	BOKEV 2A	BOKEV 2B	BOKEV 2G	BOKEV 2H

2. DETAIL OF SIDS:

No.	ROUTES	SID NAMES (CODE)			
		19L	**19R**	**01R**	**01L**
1	A791/G450/Q20	JJS 2C	JJS 2D	JJS 2E	JJS 2F
2	R460	TEPAL 2C	TEPAL 2D	TEPAL 2E	TEPAL 2F
3	W52	SEGMO 2C	SEGMO 2D	SEGMO 2E	SEGMO 2F
4	R581	DUMKA 2C	DUMKA 2D	DUMKA 2E	DUMKA 2F
5	A467	ONOTO 2C	ONOTO 2D	ONOTO 2E	ONOTO 2F
6	R472/R598	AGODA 2C	AGODA 2D	AGODA 2E	AGODA 2F
7	A462	BEMAK 2C	BEMAK 2D	BEMAK 2E	BEMAK 2F
8	B593	NOKAT 2C	NOKAT 2D	NOKAT 2E	NOKAT 2F
9	B465	SUMAG 2C	SUMAG 2D	SUMAG 2E	SUMAG 2F
10	L507	AVPOP 2C	AVPOP 2D	AVPOP 2E	AVPOP 2F
11	W112/P646	DOPID 2C	DOPID 2D	DOPID 2E	DOPID 2F
12	Q11	SURUP 2C	SURUP 2D	SURUP 2E	SURUP 2F
13	W41/A465	KAKID 2C	KAKID 2D	KAKID 2E	KAKID 2F
14	M773	BOKEV 2C	BOKEV 2D	BOKEV 2E	BOKEV 2F

3. RNAV HOLDING PROCEDURES:

3.1 RNAV holding for RNAV systems without holding functionality or for manual holding the outbound leg is 6 NM.

3.2 **PBN RNAV 1 (GNSS) HOLDING DETAILS RWY 19L / 19R:**

Taking into consideration the safety aspect during RADAR failure and subsequent transition to procedural environment, the following CONTINGENCY plan of holding shall be followed:

i. MAMRI & PALPI: ODD LEVELS

ii. BOKOT & LILDI: EVEN LEVELS

ROUTEs	NAME & COORD	I/B TRACK & LEVEL	PATTERN & SPEED	RAD/DIST FROM CEA
G450/A791/R460/ W52/Q19	MAMRI 225405.26N 0875720.05E	120 DEG M FL 70 to FL 140	LEFT 230Kts	R-297 30.5 NM
R472/R598/W69/ A467/R325	BOKOT 231315.11N 0882951.34E	185 DEG M FL 70 to FL 140	LEFT 230Kts	R-005 32.7 NM
A462/B593/B465/ L507	PALPI 225633.24N 0883620.73E	010 DEG M FL 60 to FL 140	RIGHT 210Kts	R-029 18.2 NM
A465/W41/W112/ Q10/P646	LILDI 221727.22N 0883158.16E	355 DEG M FL 70 to FL 140	RIGHT 230Kts	R-169 23.5 NM

3.3 PBN RNAV 1 (GNSS) HOLDING DETAILS RWY 01R / 01L:

Taking into consideration the safety aspect during RADAR failure and subsequent transition to procedural environment, the following CONTINGENCY plan of holding shall be followed:

i. BOVNI & GOLEL: ODD LEVELS

ii. UKBOK & KIGUP: EVEN LEVELS

ROUTEs	NAME & COORD	I/B TRACK & LEVEL	PATTERN & SPEED	RAD/DIST FROM CEA
G450/A791/Q19/ R460/W52/R325/ W69/A467	BOVNI 223942.83N 0880030.37E	090 DEG M FL 70 to FL 140	LEFT 230Kts	R-269 24.4 NM
R472/R598	UKBOK 230035.22N 0883719.16E	175 DEG M FL 80 to FL 140	LEFT 230Kts	R-026 22.2 NM
A462/B593/B465	GOLEL 222517.00N 0883155.45E	195 DEG M FL 60 to FL 140	LEFT 210Kts	R-164 15.9NM
A465/W41/Q10/M773/ W112/L507/P646	KIGUP 220717.60N 0882339.38E	015 DEG M FL 70 to FL 140	RIGHT 230Kts	R-186 33.3 NM

4. Waypoint Information

No.	Fix Identifier (Waypoint Name)	LATITUDE	LONGITUDE	CEA	
				RADIAL	DME

1	UKBOK	230035.22N	0883719.16E	026	22.2
2	KIMOM	224529.27N	0884227.10E	072	15.2
3	GOLEL	222517.00N	0883155.45E	164	15.9
4	TARUK	223747.31N	0881810.43E	253	08.5
5	BOVNI	223942.83N	0880030.37E	269	24.4
6	BOMRA	231116.16N	0875500.57E	317	42.5
7	KIDON	232218.37N	0880040.01E	331	48.1
8	UBROP	233256.75N	0881847.16E	352	52.7
9	KIGUP	220717.60N	0882339.38E	186	33.3
10	PALPI	225633.24N	0883620.73E	029	18.2
11	AKLUB	224949.81N	0884500.99E	062	19.1
12	EMSAL	223306.31N	0884125.52E	119	15.4
13	GUMAS	222724.54N	0884121.16E	135	18.7
14	GUREN	223433.57N	0881514.82E	242	12.3
15	LILDI	221727.22N	0883158.16E	169	23.5
16	IBIPO	220605.38N	0881544.68E	197	35.9
17	BOKOT	231315.11N	0882951.34E	005	32.7
18	OMBAX	231145.91N	0881442.40E	341	33.1
19	MAMRI	225405.26N	0875720.05E	297	30.5
20	CC301	230042.67N	0882908.37E	007	20.2
21	CC302	225058.77N	0883526.81E	038	13.1
22	CC303	225346.60N	0882208.03E	342	13.9
23	CC304	222252.07N	0882425.72E	188	17.8
24	CC306	222947.26N	0881755.54E	218	13.6
25	CC308	224523.69N	0881332.57E	292	13.3
26	CC312	224350.04N	0885217.88E	083	23.7
27	CC316	225855.71N	0882235.98E	348	18.7
28	CC318	224206.59N	0883434.60E	078	07.3
29	CC319	223319.45N	0881121.89E	244	16.1
30	CC321	225311.83N	0880611.09E	304	22.9
31	CC322	225106.43N	0882037.18E	332	12.0
32	CC328	223616.92N	0884450.73E	105	17.2
33	CC329	221926.61N	0883047.44E	171	21.4
34	CC330	223722.03N	0883422.16E	115	07.6

35	CC334	224226.97N	0880913.54E	277	16.4
36	CC333	222356.44N	0881649.89E	210	19.0
37	CC336	223430.44N	0884925.69E	107	21.7
38	CC337	222931.14N	0885257.59E	115	26.5
39	CC201	224948.06N	0874229.76E	283	42.1
40	CC202	225630.25N	0871622.39E	285	67.1
41	CC203	230155.51N	0872258.91E	291	62.8
42	CC204	232156.13N	0873615.25E	312	62.3
43	CC205	233007.87N	0880649.84E	340	52.8
44	CC210	221936.41N	0884833.29E	137	29.0
45	CC211	213252.95N	0881842.61E	187	67.9
46	CC212	233225.66N	0880008.18E	335	57.3
47	CC213	233842.58N	0883234.01E	006	58.2
48	CC218	221041.45N	0890018.23E	134	43.0
49	CC101	223808.27N	0862908.43E	270	108.9
50	CC102	225946.03N	0863016.74E	281	109.4
51	CC103	232910.74N	0863417.22E	296	114.6
52	CC104	234818.38N	0865155.96E	309	110.5
53	CC105	234601.59N	0872658.23E	321	85.5
54	CC106	240239.85N	0874437.86E	335	90.6
55	CC107	235916.70N	0882052.34E	357	78.7
56	CC108	211709.88N	0891551.53E	152	94.8
57	CC109	211319.73N	0882425.24E	182	87.0
58	CC110	210944.09N	0881148.47E	189	91.6
59	CC112	213524.64N	0885944.52E	155	71.7
60	SEGMO	240802N	0864551E	-	-

VECC AD 2.23 ADDITIONAL INFORMATION

I. TAXIWAYS

Designation	Width (M)	Surface	PCN	Edge Lights (EL) / Centre Line lights (C/L)	Other lights available
A (BTN BOTH RWY)	23	BITUMEN	786/F/B/W/T	C/L and EL On curves	Guard Lights & Stop Bar
A (BTN C & RWY 01L)	23	CONCRETE	1060/R/B/W/T	C/L and EL On curves	
A (BTN C & RWY 19R)	23	CONCRETE	1100/R/B/W/T	C/L and EL On curves	
B (BTN BOTH RWY)	23	BITUMEN	786/F/B/W/T	E/L	Guard Lights
B (BTN TWY A and Secondary RWY)	23	CONCRETE	1020/R/B/W/T	E/L	
C (BTN BOTH RWY)	23	BITUMEN	786/F/B/W/T	E/L	Guard Lights & Stop Bar
C (BTN A & Sec RWY)	23	CONCRETE	1100/R/B/W/T	E/L	
D	23	CONCRETE	1006/R/B/W/T	C/L	Guard Lights & Stop Bar
E	28	CONCRETE	1060/R/B/W/T	E/L	
F	23	CONCRETE	1006/R/B/W/T	E/L	
F1	23	CONCRETE	1135/R/B/W/T	C/L	Apron taxilane
F2	23	CONCRETE	1006/R/B/W/T	C/L	Apron taxilane
G	15	CONCRETE	805/R/B/W/T	NIL	
H	25	CONCRETE	1040/R/B/W/T	E/L	
H1	25	CONCRETE	1040/R/B/W/T	E/L & C/L	
J	25	CONCRETE	1040/R/B/W/T	E/L	
J1	25	CONCRETE	1040/R/B/W/T	E/L & C/L	
K	23	ASPHALT	786/F/B/W/T	E/L	Guard Lights
L	23	CONCRETE	1060/R/B/W/T	E/L & C/L	

L1	23	CONCRETE	700/R/B/W/T	E/L	
M	15	CONCRETE	1010/R/B/W/T	NIL	
R	23	CONCRETE	1010/R/B/W/T	E/L	RETIL provided.
N1	15	CONCRETE	1140/R/B/W/T	C/L	
N2	23	CONCRETE	1140/R/B/W/T	C/L	
N3	15	CONCRETE	1140/R/B/W/T	C/L	
N4	23	CONCRETE	1140/R/B/W/T	C/L	
A1	23	CONCRETE	1140/R/B/W/T	--	
B1	23	CONCRETE	1140/R/B/W/T	--	
B2	23	CONCRETE	1140/R/B/W/T	--	
N	23	CONCRETE	1240/R/A/W/T	E/L & C/L	
P	23	CONCRETE	1240/R/A/W/T	E/L & C/L on curve	Guard Lights and Stop Bar
Q1	23	CONCRETE	1260/R/A/W/T	E/L & C/L	RETIL provided.
Q2	23	CONCRETE	1260/R/A/W/T	E/L & C/L	RETIL provided.
R1	23	CONCRETE	1120/R/A/W/T	E/L & C/L	RETIL provided.

i. Taxi holding positions for CAT III ILS operations are available at 120M from the centreline of RWY 19L/01R on TWY D.

ii. The erected structure of Vidyasagar setu (second Hoogly bridge) position 223315N 0881942E with two pairs of high rise height 424 Ft AGL / 436 Ft AMSL each at two banks of the river 458 Mtrs apart. Distance / Direction / Bearing from Kolkata airport 16.5 Km/ SW/228 degree. Obstruction marked by four red coloured aviation obstruction lights and additionally four lights at intermediate level.

iii. Chimney approx 285 M(935 Ft) AMSL height at location 222750N 0880748E, not yet marked, temporarily six neon obstruction lights (CFL)provided during night. Aircraft operating in the vicinity exercise caution.

iv. Four flood lighting steel tower height 83M AMSL erected in Eden. -garden stadium position 223348N 0882044E, 14.3 Km SW from ARP Kolkata and 7850M NE from ARP Behala.

v. ASMGCS available to enhance visual surveillance at KOLKATA.

vi. Gradient Slop of Parking Stands NO. 30,31 and 32 is 0.50 %.

vii. Gradient Slop of Parking Stands NO. 22 and 25 is 1.50 %.

viii. Gradient Slop of Parking Stand NO. 27 is 1.50 %

ix. Gradient Slop of Parking Stand NO. 26is 1.50 %.

x. Gradient Slop of Parking Stands NO. 28 and 29is 1.5 %.

xi. Taxiway K available for unrestricted operation for aircraft up to Code Letter C. Fillets are provided on taxiway K at junction with Runway 19L/01R.

xii. Lock turn on Runway 19L/01R at Taxiway K intersection not permitted. Pilots are advised to make U turn abeam Taxiway K by using the fillet for reducing the runway occupancy time.

xiii. All arrival/departure to report aircraft registration on first contact with SMC on 121.9MHz alongwith other details.

xiv. The aircraft allotted stand no.15-16 and 17 will taxi in via TWY" F" and taxi out via TWY "A".

xv. VDGS, Ground marking, stand identification signs and aircraft stand manoeuvring guidance lights with stop position lights are available on stand 51, 52, 53, 54, 55, 55R, 55L, 56, 56R, 56L, 57, 57R, 57L, 58, 58R, 58L, 59, 59R, 59L, 60, 60R, 60L, 61, 61R, 61L.

xvi. For RWY 19R, Code D and Code E type of aircraft shall hold at intermediate holding position on TWY N abeam RWY 19R Glide Path.

xvii. Simultaneous operations on portion of TWY A (between TWY F and RWY 19R holding position) and TWY N4 shall not be permitted.

xviii. TWY K available up to Code C type of aircraft. Departing aircraft taxiing via TWY K for RWY 19L to hold short of TWY K on RWY 19R/01L.

xix. Three hangars for Code C aircraft, apron and link taxi track (TWY L1) available towards eastern side of primary runway.

xx. 180 degree turn not allowed for power in push back parking stand due safety concern. NSOP Aircraft need to carry their tow bar for pushback.

xxi. Grass cutting machine will be deployed on the Runway strips of RWY 19L/01R and RWY 19R/01L in coordination with ATC and will keep clear of active runway and taxiways. Pilots to exercise caution.

xxii. The paved area between parking stand 04 and 115L near Sub-Fire station has been converted into two no. Code-B temporary parking stands (Parking stand no. 02 & 03) in power in power out configuration facing north.

xxiii. Equipment Restraint Area (ERA) Markings in MARS Parking Stands is denoted by RED Colour for Code E Aircrafts and ORANGE Colour for Code C Aircrafts.

II. LOW VISIBILITY PROCEDURES

1. DEFINITIONS & ABBREVIATIONS: -

1.1 Instrument Approach Operations. An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

a) a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and

b) a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance.

Note. — Lateral and vertical navigation guidance refers to the guidance provided either by:

a) a ground-based radio navigation aid; or

b) computer-generated navigation data from ground-based, space-based, selfcontained navigation aids or a combination of these.

Instrument approach operations shall be classified based on the designed lowest operating minima below which an approach operation shall only be continued with the required visual reference as follows:

a) Type A: a minimum descent height or decision height at or above 75 m (250 ft); and

b) Type B: a decision height below 75 m (250 ft).

Type B instrument approach operations are categorized as:

1) Category I (CAT I): a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a runway visual range not less than 550 m;

2) Category II (CAT II): a decision height lower than 60 m (200 ft), but not lower than 30 m (100 ft) and a runway visual range not less than 300 m;

3) Category III (CAT III): a decision height(DH) lower than 30 m (100 ft), or no decision height and a runway visual range less than 300 m or no runway visual range limitations.

Note 1. — Where decision height (DH) and runway visual range (RVR) fall into different categories of operation, the instrument approach operation would be conducted in accordance with the requirements of the most demanding category .

Note 2. — The required visual reference means that section of the visual aids or of the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of position, in relation to the desired flight path. In the case of a circling approach operation the required visual reference is the runway environment.

Note 3. — Guidance on approach classification as it relates to instrument approach operations, procedures, runways and navigation systems is contained in the All Weather Operations Manual (Doc 9365).

1.2 Decision Height: A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.
Note. — Decision altitude (DA) is referenced to mean sea level and decision height (DH) is referenced to the threshold elevation.

1.3 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles / or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.

1.4 ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected to provide protection against interferences cause by large moving objects outside the critical area but still normally within the airfield boundary.

1.5 Low Visibility Procedures: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs.

1.6 Low Visibility Take-Off (LVTO): A term used in relation to flight operations referring to a take-off on a runway where the RVR is less than 400m.

1.7 Obstacle Free Zone: The airspace above the inner approach surface, inner transitional surface and balked landing surface and that portion of the strip bounded by these surfaces, which is not penetrated by any fixed obstacle other than of low mass and frangible mounting, required for air navigation purposes.

1.8 Runway Visual Range: The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centerline.

1.9 Safeguarding Procedures: Safeguarding Procedures (SP) are instructions for relevant airport departments and airside operators to prepare ground services and facilities for low visibility operations, in order that when LVP are implemented all SP are complete and airport is configured for Low Visibility Procedures and Low Visibility Take-offs.

1.10 Missed approach procedure: The procedure to be followed if the approach cannot be continued.

1.11 Aerodrome Operating Minima- The limits of usability of an aerodrome for:
a) take-off, expressed in terms of runway visual range and / or visibility and, if necessary, cloud conditions;
b) landing in 2 D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and
c) landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height(DA/H) appropriate to the type and/or category of the operation.

1.12 Touchdown Zone- The portion of a runway, beyond the threshold, where it is intended landing aeroplanes first contact the runway.

1.13 Visibility- Visibility for aeronautical purposes is the greater of:

a) the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;

b) The greatest distance at which lights in the vicinity of 1 000 candelas can be seen and identified against an unlit background.

Note 1. — The two distances have different values in air of a given extinction coefficient, and the latter b) varies with the background illumination. The former a) is represented by the meteorological optical range (MOR).

Note 2. — The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

1.14 The abbreviations used in descriptions of Low Visibility Procedures have the following meanings:

AFSS : AIRPORT FIRE SAFETY SERVICES
ARFF : AIRPORT RESCUE & FIRE FIGHTING
AGL : AERONAUTICAL GROUND LIGHT
AOCC : AIRPORT OPERATIONS CONTROL CENTRE
ATC : AIR TRAFFIC CONTROL
ATIS : AUTOMATIC TERMINAL INFORMATION SERVICE
FAP : FINAL APPROACH POINT
GP : GLIDE PATH
ILS : INSTRUMENT LANDING SYSTEM
IHP : INTERMEDIATE HOLDING POSITION
LOC : LOCALIZER
LSA : LOCALIZER SENSITIVE AREA
LVP : LOW VISIBILITY PROCEDURE
MET : METEOROLOGY
MID : MID POINT
OFZ : OBSTACLE FREE ZONE
RVR : RUNWAY VISUAL RANGE
SMC : SURFACE MOVEMENT CONTROL
SP : SAFE GUARDING PROCEDURES
SSO : CNS SHIFT SUPERVISORY OFFICER
TSO : TOWER SUPERVISORY OFFICER
TDZ : TOUCHDOWN ZONE
WSO : WATCH SUPERVISORY OFFICER
ASMGCS : ADVANCED SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM

2. INTRODUCTION

2.1 General

2.1.1 At NSCBI Airport, Kolkata, Runway 01R is equipped with ILS CAT-II/CATIII facilities while Runway 19L is equipped with ILS CAT II facilities. Low visibility procedures (LVP) shall be implemented at the airport before the conduct of ILS CAT-II/CAT-III Operations and/or Departures in visibility/RVR less than 800M/550M.

2.1.2 LVP shall only be implemented when Safeguarding Procedures (SP) have been completed and the airport is configured for low visibility operations.

2.1.3 The pilot shall ensure that he/she is suitably qualified and certified to carry out the required category of ILS approach. It shall be the responsibility of the Pilot to decide the category of ILS Approach he/she may wish to carry out under the given conditions

2.2 EQUIPMENT

2.2.1 The following equipment shall be serviceable to the required standard.

a) ILS localizer Transmitters (Both Main and Stand by)

b) ILS glide path Transmitter (Both Main and Stand by)

c) ILS DME or Outer Marker

d) Airfield ground lighting system (AGL) according to the requirement of Para 2.2.2.

e) Instrumented RVR system: (TDZ, MID & END for CAT III; TDZ & MID for CAT II)

f) Standby power supply for ILS

g) UPS/ Standby DG Set power for aeronautical ground lighting system.

h) Surface Movement Radar (ASMGCS (SMR) for CAT II/CAT III operations/ LVTO (for RVR below 400M).

(DGCA **CAR Section 4 Series B Part 1, Para 9.8.7.** — Surface movement radar for the maneuvering area shall be provided at an aerodrome intended for use in runway visual range conditions less than a value of 350 m.)

When any equipment and facilities listed in Para 2.2.1 above becomes unserviceable during periods of LVP, the concerned ATC unit shall advise all aircraft and accordingly suspend CAT II/CAT III/LVTO operations.
Information to this effect shall be included in ATIS broadcast.

The pilot shall ensure that he is suitably qualified and certified to carry out the required category of ILS approach.

2.2.2 The unserviceability of the above mentioned items will affect the operations as given in the table Below.:

Facility	Unserviceability	**Restrictions**
		Arrivals
ILS Localizer	One transmitter unserviceable	Suspend CAT-II / CAT-III operations
ILS Glide Path	One transmitter unserviceable	Suspend CAT-II / CAT-III operations
ILS DME	One transmitter unserviceable	Suspend CAT-II / CAT-III operations

RVR System	TDZ RVR	Suspend CAT-II / CAT-III operations
	MID RVR	Suspend CAT-II / CAT-III operations
	END RVR	Suspend CAT-III operations.
	MID and END RVR	Suspend CAT-II / CAT-III operations
Stand By Power Supply	Generator unserviceable	Suspend CAT-II / CAT-III operations
Stop Bars		Suspend CAT-III operations
Approach Lights	The inner 450 meters- more than 5 % of all lights	Suspend CAT-II / CAT-III operations
	Beyond 450 meters more than 15% of all lights	Suspend CAT-II / CAT-III operations
Runway Edge lights	More than 5% of all Lights	Suspend CAT-II / CAT-III operations
	Two adjacent lamps	Suspend CAT-II / CAT-III operations
Runway Center-line lights	More than 5% of all Lights	Suspend CAT-II / CAT-III operations
	Two adjacent lamps	Suspend CAT-II / CAT-III operations
Touchdown Zone lights	More than 10% of all Lights	Suspend CAT-II / CAT-III operations
	Two lamp in Barrette	Suspend CAT-II / CAT-III operations
Threshold lights	More than 5% of all Lights	Suspend CAT-II / CAT-III operations
	Two adjacent lamps	Suspend CAT-II / CAT-III operations
Runway End lights	More than 25% of all Lights	Suspend CAT-II / CAT-III operations
	Two adjacent lamps	Suspend CAT-II / CAT-III operations
Taxiway Centre-line lights	Not applicable to CAT-II operation	Close affected taxiways, use alternate taxi route
	Two adjacent lamps	

NOTE 1: For operations in RVR less than a value of 350 m the serviceability requirement of STOPBAR lights are:

a) no more than two lights will remain unserviceable; and

b) two adjacent lights will not remain unserviceable unless the light spacing is significantly less than that specified. (Ref Para 10.5.8, DGCA- **CAR Section 4 Series B Part 1**).

NOTE 2: For operations in RVR less than a value of 350 m, no two adjacent taxiway centre line lights be unserviceable.

NOTE 3: Un-serviceability of any of the following facilities does not affect CAT-II/CAT-III operations:

i) PAPI

ii) Taxiway edge lights on curves.

2.3 ILS SENSITIVE AND CRITICAL AREAS

2.3.1 The ILS critical and sensitive areas have been shown in **ANNEXURE C1-C4**. Protection of these areas during CAT-II/ CAT-III operations will be ensured as per Standard Operating Procedures.

Note: Signage indicating the limits of localizer and glide paths sensitive areas are provided.

2.3.2 Diagram indicating the Critical and Sensitive area of ILS for all runways shall be available with apron control, control tower and WSO.

2.4 REPORTING OF RVR VALUES

2.4.1 Touch-down zone RVR needs to be reported for Cat I operations; touch-down and mid zone RVR for Cat II operations; touch-down, mid and roll-out zone RVR for Cat III operations. In all cases, touch-down zone RVR will always be controlling RVR. However, if any other RVR is reported and is relevant (operator shall not define relevant depending on runway length/aircraft stopping distance unless approved by FSD, DGCA) it also becomes controlling. The mid zone and roll-out zone can be lower than the touch-down zone provided conditions enumerated in Note 1 below are satisfied. The following table should be used for reference;

Table 2-1

Type of Operation	RVR		
	Touch-down Zone	Mid zone	Roll-out zone
CAT I	550 m	125 m	125 m
CAT II	300 m	125 m	125 m
CAT III	175 m	125 m	125 m
CAT III (with roll-out guidance)	75/50 m	75/50 m	75/50 m

Note 1: The use of minimum RVRs in the table above is subject to:

- operator authorization;
- aeroplane authorization;
- flight crew training and qualification; and
- aerodrome facilities.

Note 2: The use of minimum RVR of 75m or 50m depends on value approved for operators and aeroplanes with roll-out guidance system.

Note 3: The values in bold font are required for the type of operation.

3. SAFEGUARDING AND LOW VISIBILITY PROCEDURE

3.1 SAFEGUARDING PROCEDURES-.

3.1.1 Criteria for Initiation of Safeguarding Procedures:

Safeguarding Procedures shall be initiated when:

a) The RVR is less than 1200 m or visibility is forecast to deteriorate to 800m or less; and/or

b) The cloud ceiling is 400 ft and forecast to fall to 200 ft or less.

3.1.2 Necessary actions to prepare the airport for Low Visibility Procedures:

a. inspection of aeronautical ground lighting system,

b. termination of all work in progress in the localizer and glide path sensitive area and the maneuvering area,

c. removal of all equipment/material from localizer and glide path sensitive area and the maneuvering area,

d. restrictions on the movement of vehicles on the maneuvering area and aprons.

3.1.3 Process of initiation of Safeguarding Procedures:

a) When meteorological conditions meet the criteria for initiation of SP as stipulated in Para 3.1.1, the Watch Supervisory Officer will inform AOCC, SSO, TSO and Duty Met Officer.

b) On receipt of the above information the concerned agencies will take action as mentioned in Para 6.

c) When all the concerned agencies have completed their necessary actions they shall report to WSO that their Safeguarding Procedure (SP) is completed and the airport is safeguarded for LVP operations.

3.2 LOW VISIBILITY PROCEDURES (LVP)

3.2.1 Criteria for implementation of Low Visibility Procedures:

Low Visibility Procedures shall be implemented when –

a) Either, TDZ, MID or END RVR is less than 800 m; and/or Cloud ceiling is less than 200 ft; and

b) Safeguarding Procedures (SP) have been completed and the airport is safeguarded.

(Note: Though LVP is implemented when RVR is less than 800m, ILS CAT I operation will continue till TDZ RVR is not less than 550m)

3.2.2 Actions to be taken upon Implementation of Low Visibility Procedures

3.2.2.1 When meteorological conditions meet the criteria for LVP as stipulated in Para 3.2.1, and airport is safeguarded, WSO shall implement Low Visibility Procedure and inform:

a) Aerodrome Tower Supervisor (TSO).

b) Approach Radar Controller (TAR)

c) Communication/Technical Shift Supervisory Officer (SSO)

d) Duty Met. Officer

3.2.2.2 Upon implementation of LVP, Tower Supervisor (TSO) will inform:

a) Fire station

b) AOCC

And ensure that "**LOW VISIBILITY PROCEDURE IN FORCE**" is included in ATIS Broadcast.

3.3 ACTIONS TO BE TAKEN BY VARIOUS AGENCIES DURING LVP AND SP ARE ENUMERATED IN ANNEXURE A.

3.4 CANCELLATIONS OF SAFEGUARDING PROCEDURES (SP) & LOW VISIBILITY PROCEDURE (LVP)

3.4.1. WSO may terminate LVP when:

a) Meteorological conditions improve and TDZ, MID & END RVR are 800 m or more and/or the cloud ceiling is 200 ft or higher, and trend is for improvement for both runways; or,

b) Facilities and equipment (listed in Para 2.2.1) necessary for CAT II/CATIII operations are degraded and/or the prevailing conditions are considered unsafe for such operations.

3.4.2. On cancellation of LVP by WSO, TSO shall inform all concerned agencies and include **"LOW VISIBILITY PROCEDURES ARE CANCELLED"** in the subsequent two ATIS broadcasts.

3.4.3. In case SP has been completed, but LVP is not initiated and subsequently meteorological conditions improve such that the visibility/RVR is more than 1200m, cloud ceiling is 400 ft or higher and both are forecast to remain above the required SP criteria, WSO may cancel SP.

4 LOW VISIBILITY OPERATING PROCEDURES

4.1 AERODROME GROUND LIGHTING

4.1.1 When Safeguarding Procedure (SP) is implemented TSO shall ensure appropriate selection of aerodrome ground lighting CAT II /CAT III facilities. These facilities shall remain selected until SP and LVP are cancelled.

4.1.2 During SP and LVP, AGL Shift Supervisor shall monitor the status of all concerned ground lights and immediately advise ATC Tower of any un-serviceability affecting the operations.

4.1.3 TSO shall report any un-serviceability of AGL observed in the Tower indicator or reported by aircraft to AOCC.

4.1.4 During the period of LVP, the lights on taxiways that are not being used may be switched off.

4.2 LIGHTING INSPECTION PROCEDURES

4.2.1. The appropriate aeronautical ground lights must be inspected during the hour preceding implementation of LVP and thereafter every subsequent two-hour period. AGL Shift Supervisor shall carry out close monitoring of serviceability of AGL through monitoring system. The lighting inspections should be accorded high priority and for this purpose aircraft operations may have to be delayed if necessary.

4.2.2. AOCC shall be responsible for organizing inspections of the relevant aeronautical ground lighting. To ensure minimum delay in completing the inspection, separate teams may inspect the landing runway, associated taxiways and apron area.

4.2.3. Only the lighting for the active runway and associated taxiways are required to be inspected.

4.3 APPROACH/ RADAR CONTROL PROCEDURES

4.3.1. During LVP the Approach Radar Controller (TAR) shall have the following information

a) Status of ILS

b) RVR information of TDZ, MID and END.

Note: Any degradation in any of the above facilities shall be immediately intimated to the arriving and departing aircraft by Approach/ Tower controller

4.3.2. In addition to the information normally transmitted by Approach Radar Controller, the information specified in 4.1.3 and 4.1.4 must be passed to the arriving aircraft on first contact or as soon as possible along-with the un-serviceability, if any, of any component parts of CAT II/CAT III facilities not previously broadcast on ATIS.

4.3.3. When reporting RVR to pilots, the TDZ RVR shall always be passed for the concerned Landing RWY.

4.3.4. In addition to Para 4.3.3

a) For CAT II Operations - If TDZ RVR is below 550m then MID RVR shall also be passed.

b) For CAT III Operations - If TDZ RVR is below 300m, MID and END RVR reading shall also be passed. TDZ, MID and END RVR shall be available.

4.3.5. Approach/Radar Controller should vector the arriving aircraft to intercept the localizer at a distance not less than 10NM from touchdown.

4.3.6. Suitable spacing between the arriving aircraft should be provided to ensure that the arriving aircraft can be given a landing clearance by 2 NM from touchdown. If there is a departure between the two arrivals, the spacing between the arriving aircraft may be suitably increased to ensure that the departing aircraft passes overhead the Localizer before the inbound aircraft reaches 2NM from touchdown

4.3.7. Approach Radar Controller shall not subject an aircraft carrying out CAT II/CAT III approaches to any speed control when within 20NM from touchdown.

4.4 AERODROME CONTROL PROCEDURES

4.4.1. Arriving aircraft shall be issued landing clearance not later than 2NM from touchdown. If landing clearance cannot be issued when the aircraft is 2NM from touchdown it shall be instructed to carry out a missed approach.

4.4.2. Arriving aircraft should be given unimpeded taxi route to allow it to clear the localizer sensitive area expeditiously.

4.4.3. The LSA in front of an arriving aircraft shall not be infringed from the time it is 2NM from the touchdown until it has completed its landing roll.

4.4.4. Landing clearance shall not be issued until: -

A preceding landing aircraft has vacated Localizer Sensitive Area (LSA) and confirmed by Pilot.

4.4.5. Aerodrome Control shall initiate emergency action if an aircraft is not seen/observed in ASMGCS/SDD/visually or not in radio contact as expected.

4.5 SURFACE MOVEMENT CONTROL PROCEDURES

4.5.1 Pilots need additional guidance and information when taxiing during periods of reduced visibility. The view from the cockpit of the aircraft is very limited. Therefore, taxi instructions and essential traffic information should be passed in a clear and concise manner.

4.5.2 SMC shall inform all taxiing aircraft of the preceding taxiing or holding aircraft.

4.5.3 Aircraft should be assigned taxi-routes in accordance with Tables 4-1 and 4-2, as far as practicable. However, due to infrastructural constraints at NSCBI Airport, Kolkata, there are some conflicting taxiway intersections as well as overlapping portions in arrival and departure taxi routes in the mentioned tables. Sometimes, it may be necessary to route aircraft via alternative taxiways due to operational reasons. However, alternate taxi-route will be exercised during CAT-II conditions only.

4.5.4 The following precautions will be observed in conflicting taxi areas:

a. Follow-me service

b. Intermediate Holding Position Markings and Lightings

4.5.5 Surface Movement Controller should monitor the progress of arriving aircraft on ASMGCS Display as they vacate the runway after landing and ensure that they do not stop within the Localizer Sensitive Area [LSA] thereby degrading ILS integrity for subsequent landing aircraft. Pilots shall report runway vacated on RTF using the guidance of color coded exit taxiway centerline lights.

4.5.6 Vehicle movement, when RVR is less than 550m, should be restricted. Only operationally essential vehicle duly authorized by Apron control should be permitted to operate on the maneuvering area. These vehicles shall remain outside the Localizer Sensitive Area(LSA). Any movement of vehicle on the maneuvering area shall be coordinated with ATC. During CAT II/CATIII operations, vehicle fitted with transponder (vehicle locators) should only be permitted on the maneuvering area.

4.6 LOW VISIBILITY PROCEDURE TAXI ROUTE

4.6.1 After LVP is declared, AGL Shift Supervisor will ensure that the lead-in lights of all CATII/III compatible stands are switched ON.

4.6.2 During CAT II or CAT III ILS visibility conditions i.e. when RVR reduced to less than 550 m, ‘Follow me’ service shall be provided whenever deemed necessary by ATC or requested by pilot.“FOLLOW ME” service can be put into use up to a minimum visibility of 50m.

4.6.3 Person providing “FOLLOW ME” service shall be trained and fully familiar with the taxi routes intersections and other maneuvering area/apron/parking stands.

4.6.4 During CAT II/CAT III ILS operations, aircraft should be routed in accordance with the pre-designated taxi routes as given in Table 4-1 and Table 4-2 below, unless otherwise necessitated due operational requirements:

Table 4-1

CAT II, CAT III RWY 01R		
RWY 01R	Arrival taxi routes to Apron2 & Apron3	Route I: Vacate via TWY Q2, cross RWY 01L, N, N2, F and then via: i. F1 to stand 48 to 57R (Apron2) ii. F2 to stand 57 to 61L (Apron3) OR Route II: Vacate via TWY P, taxi via N, N2, F and then via: i. F1 to stand 48 to 57R (Apron2) ii. F2 to stand 57 to 61L (Apron3) OR Route III: Vacate via TWY A, Cross RWY 01L/ 19R, taxi via A, N4, A, J1 and then via: i. F1 to stand 48 to 57R (Apron2) ii. F2 to stand 57 to 61L (Apron3)
	Departure taxi routes from Apron2 & Apron3	Taxi to CAT III holding point RWY 01R via i. Stands 48 to 57R (Apron2) – F1, H1, A, D ii. Stands 57 to 61L (Apron3) – F2, H1, A, D
	Hotspots	i. Junction of F1, F2, H1, J1(during route I, II and III) ii. Junction of A, H, H1, A((during route III) iii. Junction of A, D, A(during route III)
	Bidirectional taxi routes	i. Portion of A between D & H1 (during route III) ii. F1 iii. F2

Table 4-2

CAT II, , CAT III RWY 19L

RWY 19L	Arrival taxi routes to Apron2 & Apron3	Vacate via TWY D, cross 19R/01L taxi via A, H1, and then via: i. F1 to stand 48 to 57R (Apron2) ii. F2 to stand 57 to 61L (Apron3)
	Departure taxi routes from Apron2 & Apron3	**Route I:** Taxi to CAT III holding position RWY 19L-P intersection departure via i. Stands 48 to 57R (Apron2) - F1, F, N2, N, P ii. Stands 57 to 61L(Apron3)– F2, F1, F, N2, N, P OR **Route II**: Taxi to CAT III holding position RWY 19L-A intersection departure, via i. Stands 48 to 57R (Apron2) – F1, F, N4, A ii. Stands 57 to 61L(Apron3)– F2, F1, F, N4, A **Note:** Aircraft may be permitted to enter Runway 19L via taxiway A, when pilot informs about his intention to depart from 'A' intersection or when taxiway P is not available.
	Hotspot	Junction of F1, F2.
	Bidirectional taxi routes	i. F1 ii. F2

Note 1: In contact stands on Apron 2 and Apron 3 shall be used during LVP.

Note 2 : Airline operators will be responsible for ensuring that the parking stand area is clear of all equipment when aircraft is taxiing in for parking (airlines may consider engaging wing walkers for the same.)

Note 3: The airline operators will ensure that push-back area is clear of all equipment before push back is commenced.

Note 4: In case parking stands on Apron 2 & 3 are not available, Apron 1 & Apron 4 may be utilized by providing follow-me jeep.

4.7 VEHICULAR MOVEMENT

4.7.1 Vehicle movement, when RVR is less than 550m, should be restricted. Only operationally essential vehicle duly authorized by Apron Control should be permitted to operate on the maneuvering area. These vehicles shall remain outside the Localizer Sensitive Area [LSA].

4.7.2 Any movement of vehicle on the maneuvering area shall be coordinated with ATC.

4.7.3 During CAT II/ CAT III operations, vehicles fitted with squids shall only be permitted on the maneuvering area. However, other vehicles crossing taxiways, on service road and Apron taxiways shall be regulated by AOCC.

4.7.4 A diagram showing vehicular lanes is provided in ANNEXURE B.

4.8 AIRCRAFT MOVEMENT

4.8.1 Aircraft shall not be held at any point closer to the runway than the CAT-II/CATIII holding point/ stop-bar.

4.8.2 If any portion of taxiways mentioned in Table 4-1 is not available for taxiing, CAT-II/CATIII operation Runway 01R shall be suspended.

4.8.3 If any portion of taxi routes mentioned in Table 4-2 is not available for taxiing, CATII operation Runway19L shall be suspended.

4.9 LOW VISIBILITY TAKE-OFF (LVTO)

4.9.1 LVTO pertains to take off when the RVR is below 400 m and is applicable whenever the reported RVR in any zone (touch-down/mid/roll-out RVR) is below 400 m. The facilities and conditions of Table 4-3 below will be as per the lowest RVR reported in any zone (e.g. if the RVR is 400/300/300 representing the three zones, then the 300 m will be the RVR for reckoning facilities and conditions of Table 4-3. If the RVR is 300/150/Not Reported, then 150 m will be the reckoning RVR and as the RVR is below 200 m all three RVRs are required)

Table 4-3

Take-Off RVR/Visibility	
Facilities	**RVR/Visibility[1] CAT A, B,C & D**
Adequate visual reference[2] (Day only)	500 m
Runway Edge Lights or Runway Center Line Markings[3]	400 m
Runway Edge Lights and Runway Center Line Markings[3]	300 m
Runway Edge Lights and Runway Center Line Lights	200 m

Runway Edge Lights and Runway Center Line Lights and relevant RVR information[4]	150 m
High Intensity Runway Edge Lights and Runway Center Line Lights (Spacing 15 m or less) and relevant RVR information[4]	125 m
High Intensity Runway Edge Lights and Runway Center Line Lights (Spacing 15 m or less), approved lateral guidance system and relevant RVR information[4]	75 m

Note 1 – The TDZ RVR/VIS may be assessed by the pilot

Note 2 - Adequate Visual reference means, that a pilot is able to continuously identify the take-off surface and maintain directional control.

Note 3 - For night operations at least runway edge lights or center line lights and runway end lights are available.

Note 4 - The required RVR must be achieved for all relevant RVR reporting points (touchdown, mid- point and stop-end/roll-out). The governing RVR shall be the lowest of the reported RVRs.

4.9.2 An operator shall not conduct low visibility take-offs in less than 400 m RVR unless approved by DGCA.

4.9.3 An operator shall not conduct take-off with visibility/RVR less than Category I conditions unless low visibility procedures are enforced.

5. DESCRIPTION OF EQUIPMENT AND SERVICES

5.1 RUNWAY VISUAL RANGE (RVR)

5.1.1 There are three transmissometers each for RWY 19L and RWY 01R to record RVR values. One transmissometer is located each at touchdown zone, runway midpoint, and end of runway. RVR values always refer to Touchdown RVR (TDZ), Mid-point RVR (MID) and Stop End RVR (END).

5.1.2 RVR is reported in the following scales:

a) From 2000m to 1200m, increments of 100m
b) From 1200m to 800m, increments of 50m
c) From 800m to 50m, increments of 25m

5.1.3 Equipment serviceability for CAT II/CAT III operations:

a) For CAT II operations TDZ and MID RVR shall be available.
b) For CAT III operations TDZ, MID and END RVR shall be available.

5.2 AERONAUTICAL GROUND LIGHTING(AGL) SYSTEM

5.2.1 The Precision Approach lighting system for CAT II/CAT III operations are installed for RWY 01R and that for CAT II/ CAT III operations are installed for RWY 19L at NSCBI Airport.

5.2.2 The Electrical Power Supply Systems for AGL are so designed that the time interval between failure of the primary source of power and the complete restoration of the services for visual aids associated CATII/ CATIII operations, shall be in accordance with maximum switch-over times in Table 8-1 of Para 8 of DGCA CAR, Section 4 Series B Part I.

5.2.3 During the LVP operation, irrespective of RVRs, the state electricity board (CESC) power supply will be the primary source through on line UPS for the systems which require a maximum of one second switchover time and the DG set shall be put into operation mode to meet the requirements in case of power failure. The power change over for other systems which are not covered with the UPS back up shall be getting the power supply through DG Sets within 15 seconds from the time of power failure. For the systems covered with UPS back up, the changeover will be zero seconds.

5.2.4 Stop Bars: Stop bars have been provided as follows:

a)On taxiway D, at CAT II/CAT III holding position for runway 01R.
b)On taxiway C, at CAT II holding position for runway 01R.
c)On taxiway A, at CAT III holding position for runway 19L.
d)On taxiway P, at CAT III holding position for runway 19L

5.2.5 CAT-III taxiway centerline lights: Taxiways P,N,N2,N4, A, D, F,F1, F2, H1, J1, have been provided with CAT-III taxiway centerline lights.

5.2.6 Entry taxiway lights: The Stop Bars mentioned in Para 5.2.4. have been provided with interlocked Entry taxiway centerline lights.

5.2.7 Exit taxiway lights: The alternate yellow and green centerline lights for exit taxiways have been installed on taxiway A,Q2,P for CAT III operation runway 01R and TWY D for CAT III operation runway 19L.

5.2.8 Taxiway edge lights: Taxiway edge lights are available on all Taxiways except TWY-N1, N2, N3, N4, P on straight, G, M, F, F1 and F2.

5.2.9 Runway Turn pad lights: Lighted guidance (Turn Pad Lighting) is available for making180 degree turn in the beginning of RWY 19L.

5.2.10 Runway Guard lights: Runway Guard lights have been provided at CAT I holding positions on taxiways A, B, C,D,P and K.

5.2.11 Parking Stands: The following parking stands have been provided with the CAT III stand manoeuvring Centerline lighting system:

a) Parking stand 48, 49, 50, 51, 52, 53, 54, 55L, 55, 55R, 56L, 56, 56R,57R in Apron-2

b) Parking stand 57, 57L, 58L, 58, 58R, 59L, 59, 59R, 60L, 60, 60R, 61L, 61 and 61R in Apron-3

Note: Parking stand manoeuvring guidance lights are not available in following stands:

a) On Apron 1: 04 to 11L and 15 to 20.

b) On Apron 2: 21 to 25 and 27 to 32.

c) On Apron 3: 33 to 40.

d) On Apron 4: 101 to 115L.

5.2.12 Intermediate Holding Position (IHP) lights: Intermediate Holding Position (IHP) lights have been provided on the following taxiways:

i. F1 Short of junction of F1, F2, H1, J1

ii. F2 Short of junction of F1, F2, H1, J1

iii. H1 Short of junction of F1, F2, H1, J1

iv. J1 Short of junction of F1, F2, H1, J1

v. A North of junction of A & D

vi. A South of junction of A & D

vii. A North of junction of A, H & H1

viii. A South of junction of A & H1

ix. A Short of 19R/01L near Cat II holding position

x. N1 South of junction of N & N1, N2, N3, N4

xi. N2 South of junction of N & N1, N2, N3, N4

xii. N3 South of junction of N & N1, N2, N3, N4

xiii. N4 South of junction of N & N1, N2, N3, N4

xiv. N Short of 19R GP antenna on the southern side

xv. P North of junction of P & N

5.3 NON-VISUAL GROUND SURVEILLANCE SYSTEM:

5.3.1 NSCBI Airport has been equipped with (Advanced Surface Movement Guidance and Control System) ASMGCS, which has been approved for level 1 operations. The system provides non-visual electronics surveillance of manoeuvring area and facilitates the controllers to identify potential ground conflict and runway incursions.

5.3.2 For CAT III and CAT II operations below 350m availability of NonVisual Surveillance System such as ASMGCS is mandatory.

5.4 NAVIGATIONAL AIDS

5.4.1 RWY 01R has been equipped with Instrument Landing System (ILS) for CATII/CATIII operations while RWY 19L has been equipped with Instrument Landing System (ILS) for CATII operations.

5.4.2 The ILS Category Monitor Panel at the Control Tower and Approach Radar consoles indicate availability of ILS category by displaying the status of the following equipment:

i. Localizer transmitters

ii. Glide path transmitters

iii. ILS DME

Note: Since indicators in Tower and Approach Radar display status of only the operating localizer and Glide-path transmitters, SSO shall promptly inform WSO and TSO about failure of any of main and/or standby transmitters of any unit, even if nonoperative.

5.4.3 ILS equipment serviceability required for CAT II/CAT III operations: -

i. Both main and standby localizer transmitters;

ii. Both main and standby glide path transmitters;

iii. Main and Standby power supply for each unit of ILS.

iv. ILS DME

5.5 AIRPORT RESCUE & FIRE FIGHTING SERVICE (ARFF)

5.5.1 During LVP following predetermined positions (PDP) will be taken by ARFF vehicles:

a) Fire Emergency Road north of TWY-L

b) In front of main fire station

c) Near taxiway B (on old fire approach road between TWY-A and Secondary RWY)

d) Near 19R GP

5.5.3 ARFF vehicles shall move from PDPs only after positive clearance from ATC.

5.5.4 In the event of an incident when LVP are in force, Aerodrome Control and SMC should provide the maximum assistance in directing ARFF to required location.

ANNEXURE "A"
ACTIONS TO BE TAKEN BY VARIOUS AGENCIES

6.1 Before commencement of winter season, a meeting will be held by Airport Director NSCBI Airport, AAI along with all stake-holders, in the month of November every year to inform all airlines and agencies operating at airport about their roles/ responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

6.2 All the agencies shall ensure that staff and drivers are suitably trained during LVP operations.

6.3 **ACTION BY WATCH SUPERVISORY OFFICER (WSO), AAI**

6.3.1 Implementation of Safeguarding Procedures - When RVR is less than 1200m and visibility is forecast to deteriorate 800m or below and/or the cloud ceiling is 400ft and is forecast to fall to 200ft or less, WSO shall initiate Safeguarding Procedures and inform:

a) AOCC

b) Communication/ Technical supervisor (SSO)

c) Tower Supervisor (TSO).

d) Duty Met Officer

6.3.2 Implementation of LVP

6.3.2.1 WSO shall implement Low Visibility Procedures when either

a) TDZ, MID or END RVR is less than 800m; and/or Cloud ceiling is less than 200ft.

and

b) Safeguarding Procedures (SP) have been completed and the airport is configured for Low Visibility Operations.

6.3.2.2 WSO shall inform:

a) Aerodrome Tower Supervisor (TSO).

b) Approach Radar Controller (TAR)

c) Communication/Technical Shift Supervisory Officer (SSO)

d) Duty Met. Officer

6.3.3 Termination of LVP/ SP:

6.3.3.1. WSO may terminate LVP when;

a) Meteorological conditions improve and TDZ, MID & END RVR are 800 m or moreand the cloud ceiling is 200 ft or higher, and trend is for improvement for both runways, or

b) Facilities, equipment and services necessary for CAT II/CAT III operations are degraded and/or the prevailing conditions are considered unsafe for such operations.

6.3.3.2. WSO should intimate cancellation of LVP to:

a) Aerodrome Tower Supervisor (TSO).

b) Approach Radar Controller (TAR)

c) Communication/Technical Shift Supervisory Officer (SSO)

6.3.3.3. In case SP has been completed, but LVP is not initiated and subsequently meteorological conditions improve such that the visibility/RVR is more than 1200m, cloud ceiling is 400 ft or higher and both are forecast to remain above the required SP criteria, WSO may cancel SP.

6.4 <u>ACTION BY TOWER SUPERVISOR:</u>

6.4.1 On being notified by WSO that Safeguarding Procedures have been initiated, TSO shall:

a) inform Aerodrome Rescue & Fire Fighting Services,

b) check ILS status

c) check that AGL is correctly selected and operating properly

d) check RVR displays

e) check ASMGCS status (for operations below RVR 350 m)

6.4.2 On commencement of Low Visibility Procedures, TSO shall:

a) inform Aerodrome Rescue & Fire Fighting Services & AOCC

b) Check ATIS broadcast and include the message that “ILS CAT II/CAT III Low Visibility Procedures in Force”.

c) Ensure that record of all actions pertaining to ATC is maintained with time and signed by the officer taking action.

6.4.3 TSO shall promptly inform SSO and/or AOCC, as appropriate, of any unserviceability of ILS/ AGL observed or reported.

A-CDM application On cancellation of LVP by WSO, TSO shall inform all concerned agencies and include **“LOW VISIBILITY PROCEDURES ARE CANCELLED”** in the subsequent two ATIS broadcasts.

6.5 <u>ACTION BY TOWER CONTROLLER</u>

6.5.1 After the commencement of Low Visibility Procedures, Tower Controller shall:

a) Give landing clearance to aircraft not later than 2NM from touchdown.

b) Inform changes in RVR readings to the landing aircraft.

c) Give an unimpeded taxi route to arriving aircraft to allow it to clear the Localizer Sensitive Area expeditiously.

d) Inform pilots about failures of ILS, lighting system, transmissometer relevant to ILS CAT II/ CAT III Low Visibility Operations.

e) Initiate emergency action if aircraft on CAT II/ CAT III ILS is not seen (on radar display or otherwise) or not in radio contact, as expected.

6.6 ACTION BY SURFACE MOVEMENT CONTROLLER

6.6.1 During the period the Low Visibility Procedures are effective the Surface Movement Controller shall:

a) Monitor all surface movement of aircraft and vehicles on the manoeuvring area.

b) Inform all taxiing aircraft of the preceding taxiing or holding aircraft.

c) Hand over only one aircraft at a time to Tower Controller.

d) Permit only the vehicles equipped with squids in the manoeuvring area during CAT III operations. However, other vehicles crossing taxiway-service road crossings, shall be regulated by official of Airside Operations by deploying DGR manpower.

6.6.2 Pilots need additional guidance and information when taxiing during periods of reduced visibility. The view from the cockpit of the aircraft is very limited. Therefore, taxi instructions and essential traffic information should be passed in a clear and concise manner.

6.6.3 Aircraft should be assigned taxi-routes in accordance Tables 4-1 & 4-2, as far as practicable. However, due to infrastructural constraints at NSCBI Airport, Kolkata, there are some conflicting taxiway intersections as well as overlapping portions in arrival and departure taxi routes in Table 4-1& 4-2. Sometimes, it may be necessary to route aircraft via alternative taxiways due to operational reasons.

6.7 ACTION BY APPROACH/RADAR CONTROLLER

6.7.1 On being advised by WSO that ILS CAT II/CAT III Low Visibility Procedures are in force, the Approach/Radar Controller shall: -

a) Inform the arriving aircraft **"ILS CAT II/CAT III "Low Visibility Procedures in Force".**

(**Note:** Ensure that Pilot acknowledges of being cleared for ILS CAT II/CAT III approach.)

b) Inform TDZ RVR to arriving aircraft and in addition:

(i) For CAT II operations - If TDZ RVR is below 550m then MID RVR shall also be passed.

(ii) For CAT III operations – If TDZ is below 300m, then MID & End RVR readings shall also be passed.

NOTE: After an aircraft is 8NM from Touch Down or has passed outer marker, RVR observations need not be passed unless there is changes in RVR values.

c) Vector the aircraft to intercept the localizer not less than 10NM from touchdown.

d) Not subject an aircraft to any speed control when within 20NM from touchdown.

e) Provide suitable spacing so as to ensure that landing clearance can be issued to arriving aircraft not later than 2NM from touchdown.

6.8 ACTION BY CNS SHIFT SUPERVISORY OFFICER (SSO)

6.8.1 On receipt of "Outlook for LVP" from the WSO/TSO, SSO shall:

a) inform the Duty Officer, Equipment Room

b) check the status of:

i. Main and standby ILS system (LLZ/Glide Path/ILS-DME); and

ii. Indicators in the ATC units.

iii. ASMGCS

c) Inform WSO/Tower Supervisor of any un-serviceability in the equipment which is likely to affect ILS CAT-II/CAT-III operation.

6.8.2 On receipt of "Advisory Message" from WSO that LVP are to be made effective SSO shall maintain continuous monitor the performance of ILS equipment and will inform WSO of any un-serviceability which may affect ILS CAT II/ CAT III operation.

6.9 ACTION BY DUTY OFFICER, METEOROLOGICAL OFFICE

6.9.1 Duty Met Officer shall issue an 'Outlook for Low Visibility Procedures' to the Watch Supervisory Officer [WSO] whenever RVR Runway 19L/01R is 1200m or less and visibility/RVR is forecast to deteriorate to 800 meters or less and/or cloud ceiling is 400 ft or less and expected to fall to 200 feet or less.

6.9.2 Whenever RVR Runway 19L/01R is likely to fall below 800 meters and/or cloud ceiling is likely to fall to 200 feet or less within next two hours, the Duty Met Officer shall issue an 'Advisory Message' to WSO. (All the relevant transmissometer are serviceable.

6.9.3 Whenever the RVR and/or cloud ceiling are 800 meter or more and/or 200 feet or more respectively with trend towards improvement, the Duty Met Officer shall advise WSO about such improving weather conditions for the purpose of termination of LVP operation.

6.9.4 The Duty Met Officer shall ensure that the all the Met equipment including RVR displays in ATC units in the Control Tower and Approach Control are serviceable. He shall also ensure that RVR/visibility recorders of Touch down zone, Mid-Point and Roll-out positions are serviceable.

NOTE : Due to high variability of meteorological elements in space and time and the limitations of forecasting techniques available, it may not be always possible to issue a precise forecast of RVR particularly in case of transient weather phenomenon within two hours.

6.10 ACTION BY AOCC:

6.10.1 ACTION BY DUTY MANAGER, APRON CONTROL/ AOCC

6.10.1.1 On receipt of instruction regarding implementing Safeguarding Procedure (SP) from ATC, duty Manager Apron Control/AOCC shall initiate following actions:

a. Inform all users of apron (airlines/stakeholders) regarding initiation of SP through email by using the following text: *"SAFEGUARDING PROCEDURE IMPLEMENTATION IN PROGRESS".*

b. Restrict vehicular traffic on movement area. This shall be achieved by advising CISF Control Room (033-25691290) to stop entry of vehicles through vehicle entry gate nos. 01, 05 and 10.

c. Close the available gates/barriers on perimeter road leading to critical / sensitive area of navigational aids and active RWY:-

i. Gate opposite watch tower no. 04

ii. Gate opposite watch tower no. 07

iii. Gate near glide path RWY 19L (opposite watch tower no. 11)

iv. Gate connecting fire emergency road with perimeter road.

v. Service road connecting the beginning of Runway 19L and 01R.

vi. Provide temporary barriers to stop the vehicular traffic on outer perimeter road:

vii. Near watch tower no. 02.

d. Temporary barriers to stop the' movement of vehicular traffic on Apron (where vehicular lane crossing taxiways) be provided at the following locations:

i. On vehicular lane junction opposite Air India hanger.

ii. On vehicular lane adjacent to parking stand no. 6l.

iii. On vehicular lane in between parking stand no. 56 and 57;

iv. On vehicular lane leading to Apron (in between stand no, 111 and 48),

v. On vehicular lane leading to Apron (in between stand no. 07L and 08).

e. Vehicles fitted with Squid shall only be permitted to operate on maneuvering area.

f. Inform AGL shift supervisor over walkie-talkie to check and confirm serviceability/availability of UPS & Generator and keep the DG Set in to Auto operation mode.

g. Inform Civil and Electrical Maintenance Division to suspend all works in progress in operational area and removal of men, machinery and material from the work site, if in progress.

h. Inspect following facilities with AGL Shift supervisor to ensure serviceability / availability:-

i. RWY 01R ALS (CAT-III),

ii. Runway edge lights,

iii. Runway threshold and end lights,

iv. Runway Centre line lights,

v. Runway touchdown lights,

vi. Runway guard light and stop bar lights,

vii. Taxiway edge lights and taxiway centre line lights,

viii. Taxi-holding position lights

i. Inform WSO/TSO regarding completion of Safeguarding Procedures

j. Inform all users of Apron (airlines/stakeholders) regarding completion of safeguard procedure and/or enforcement of Low Visibility Procedure through e-mail by using the following text-

"*Safeguard Procedure Implemented*", and/ or "*Low Visibility Procedure Enforced*" respectively and thereafter confirm compliance to ATC Tower.

k. In case of continuous departure (gap between the departures is 10 minutes or less) or otherwise, as and when aircraft is pushed back for departure under LVP, ensure that AGL is running on UPS and Genset in Auto Mode.

l. Switch-off secondary power supply in co-ordination with ATC Tower if there is no arrival and/or departure

m. Ensure availability of 'Follow Me' service to aircraft as and when requested by ATC.

n. No vehicle on Apron shall enter / cross in the vicinity of runway or any taxiway without permission from ATC tower.

o. Positive co-ordination to be ensured with ATC tower, if any vehicle of Fire Services, Civil/Electrical Division or any other agency has to enter the runway or taxiways for urgent operational requirement.

p. As and when ATC notify tor withdrawal of Safeguard Procedure, all concerned may be intimated through e-mail by using the following text.

"*SAFEGUARD PROCEDURE WITHDRAWAL IN PROGRESS*"

q. When Safeguard Procedure is withdrawn, cancellation of LVP shall be intimated to all concerned through e-mail by using the following text –

"***SAFEGUARD PROCEDURE WITHDRAWN AND LOW VISIBILITY PROCEDURE CANCELLED***"

and confirm compliance to ATC Tower.

6.11 ACTION BY SHIFT ENGINEERS (ELECTRICAL)

6.11.1 On receipt of notification to implement Safeguarding Procedures / Low Visibility Procedures from AOCC/Apron Control, AGL Shift Supervisor will initiate following actions:

a. Check in co-ordination with Apron Control that following visual aids associated with RWY 01R/ RWY 19L are serviceable and can be operated at full intensity.

i. RWY 01R ALS (CAT-III)/ RWY 19L ALS (CAT III),

ii. Runway edge lights,

iii. Runway threshold and end lights,

iv. Runway Centre line lights,

v. Runway touchdown lights,

vi. Runway guard light and stop bar lights,

vii. Taxiway edge lights and taxiway centre line lights,

viii.Intermediate -holding position lights,

Note: All the AGL circuits shall be on remote with absolute control authority with TSO

b. Inform the serviceability of above visual lighting aids to Airport Manager, AOCC.

c. Ensure that AFL Sub Station and CCR Hall is manned and position himself at CCR Hall AGL SCADA Monitoring Station and ensure the power supply arrangements as specified at Para 5.2.3 above.

d. Inform the un-serviceability or any change in status of any facility/systems to Airport Manager, Apron Control/AOCC immediately.

e. Ensure that no electrical maintenance works is carried out during LVP either in power house or on any other electrical facilities used during CAT II/CAT III operations

f. Maintain a constant listening watch on R/T **146.2 MHz .**

6.12 ACTION BY SHIFT ENGINEERS (CIVIL)

6.12.1 On receipt of notification to implement Safeguarding procedure / Low Visibility Procedures from AOCC/Apron Control, Shift Engineer, Civil Section will initiate following actions and confirm to AOCC/apron Control.

a. Stop the work, if in progress, in the operational area.

b. Remove men, machinery and material from the work site and ensure that they should leave the operational area.

c. Resuming of work shall only be initiated after obtaining positive clearance from Apron Control/ AOCC.

6.13 ACTION BY AIRPORT MANAGER (DEPARTURE)

6.13.1 The Airport Manager (departure) will be overall responsible for ensuring smooth coordination between all AAI units and other concerned agencies. He will also ensure that all entry/exit through NTS gates to airside are restricted and all gates shall remain closed during LVP

6.14 ACTION BY AIRPORT RESCUE & FIRE FIGHTING SERVICE (ARFFS)

6.14.1 On receipt of notification of implementation of safeguarding procedures/LVP, Airport Rescue arid Fire Fighting Service (ARFFS) will proceed to Pre-determined Positions (PDPs) after obtaining positive clearance from ATC prior before entering maneuvering area. Aerodrome Control and SMC should provide the maximum assistance in directing ARFFS to required location

6.15 ACTION BY AIRPORT SECURITY POLICE (CISF)

6.15.1 On receipt of notification from Apron Control that safeguarding procedure is being implemented, CISF Control Room shall initiate following actions:

a. Inform all access gates leading to operational area and CISF Posts under their respective controls in operational area to STOP entry of vehicles/personnel.

b. Confirm compliance on above to Apron Control.

c. Only essential vehicles (e.g. catering, refueling or any other vehicle authorized by Apron Manager) may be permitted to use service road.

d. For carrying out security checks of CISF security personnel, security vehicles will be escorted by Follow-me service in co-ordination with Apron Control.

e. The Inspector in-charge shall ensure that movement of security personnel is restricted through NTB gate only Officer shall ensure by deploying adequate manpower that power house and other vital electrical installations are properly secured and protected against any unauthorized intrusion.

6.16 ACTION BY OTHER AIRPORT AGENCIES (AIRLINES, REFUELING COMPANIES, CATERING AGENCIES, AIRPORT POLICE, CUSTOMS, IMMIGRATION, HEALTH ETC.) :

6.16.1 All agencies operating in the operational area shall ensure that minimum number of their vehicles, as are absolutely essential for aircraft operations, operate in the operational area. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.

6.16.2 All the vehicles must have their obstruction lights "ON" during operation of low visibility procedures.

6.16.3 Follow all instructions/sign boards provided for vehicular movement area/service roads.

6.16.4 No vehicle/equipment/personnel shall enter in and around the vicinity of the runways or taxi-tracks except with prior permission of AAI Apron Control who in turn shall coordinate with aerodrome control tower.

ANNEXURE-B: DIAGRAM SHOWING VEHICULAR LANES (APRON 1, 2 & 3)

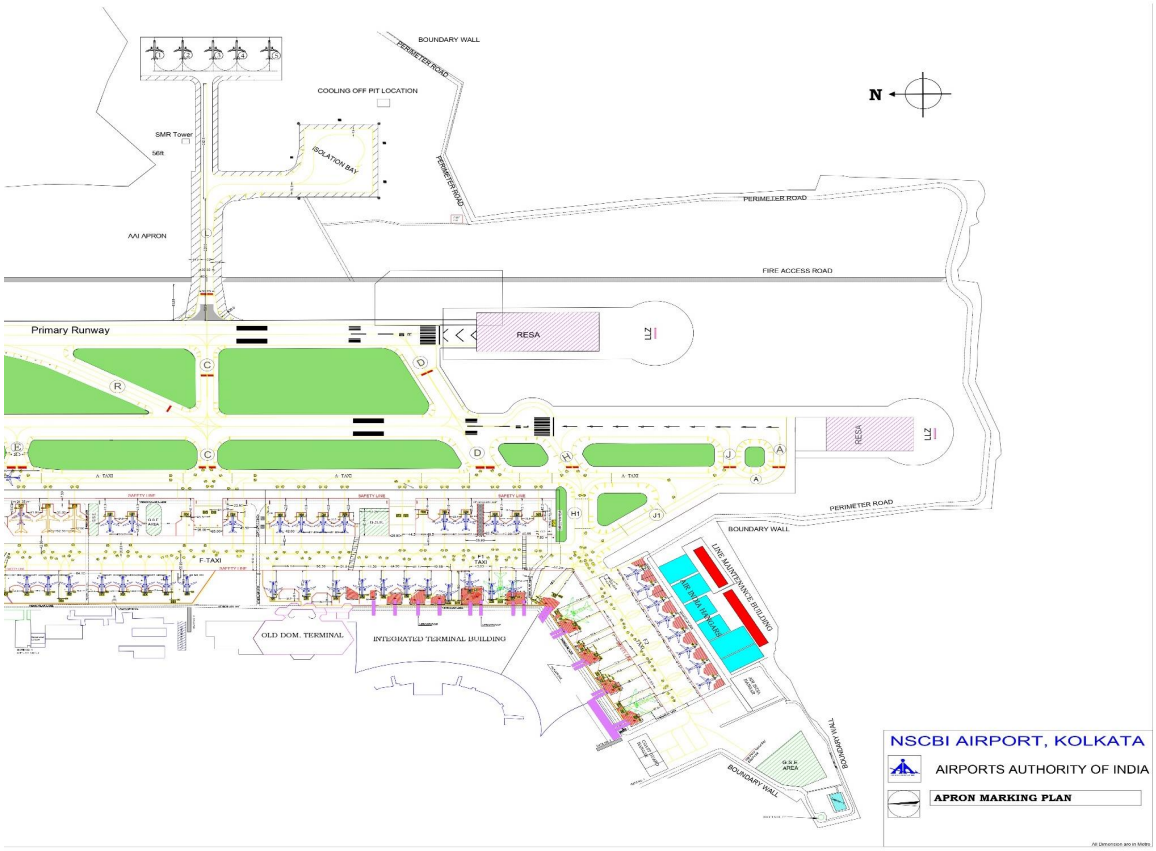
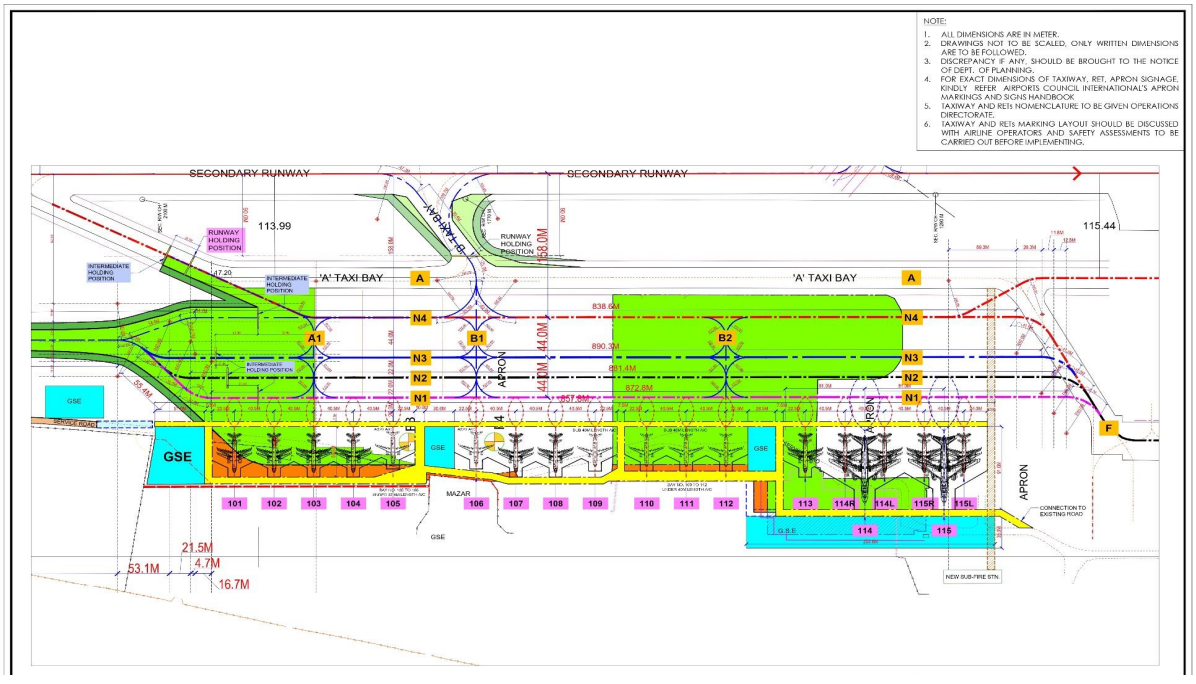
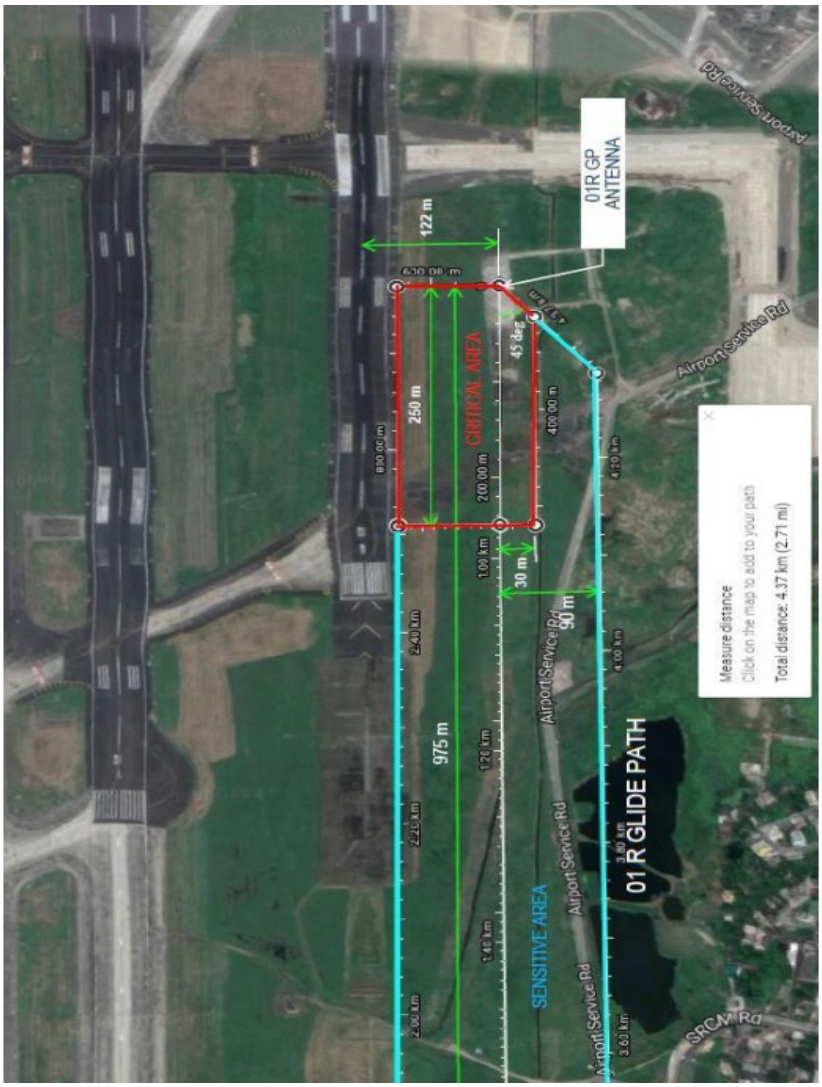


DIAGRAM SHOWING VEHICULAR LANES (APRON 4)

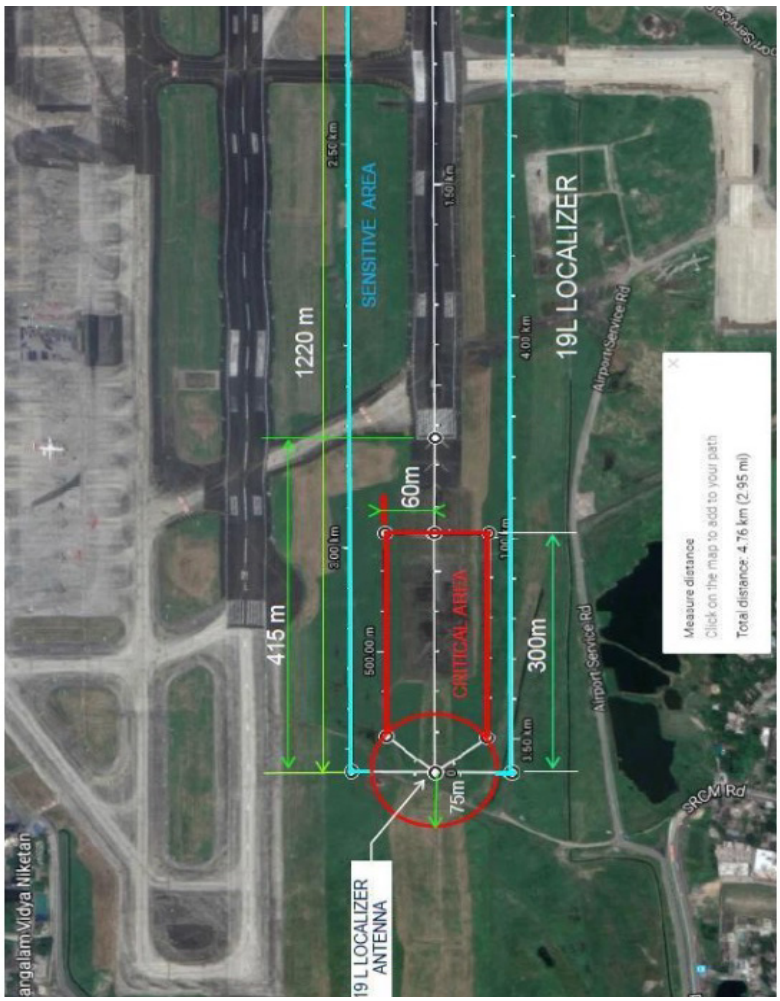



ritical and Sensitive Area for Localizer Runway 01R
01 R LOCALIZER
2750 m
785 m
90 m
120 m
90 m
Airport Service Rd
Hyundai Service Center
Sri Aurobindo Shiksha Sadan
Ganganagar Haat
Old Jessore Rd
Sir Ramesh Mitra Rd
Measure distance
Click on the map to add to your path
Total distance: 9.96 km (6.19 mi)

Critical and Sensitive Area for Glide-path Runway 01R



cal and Sensitive Area for Localizer Runway 19L



tical and Sensitive Area Glide-path Runway 19L



III TRANSPONDER OPERATING PROCEDURES

For Departure:

1.At The Gate/Stand: Select Stby and enter the Discrete SSR Code received from Clearance Delivery and the three letter ICAO Designator followed by the Flight Identification Number through the FMS or the Transponder Control Panel,depending on the Airborne Equipment.

2.On requesting Pushback/Taxi (Whichever is earlier): Select Transponder or equivalent and auto if available

3.When Lining Up: Select TCAS only after receiving the clearance to Line Up

For Arrival:

1.When on the Runway keep TCAS(TA/RA) selected

2.After vacating the Runway: Select Transponder or equivalent and Auto if available TCAS shall be deselected when vacating the Runway.

3.Parked on Stand: Select Stby:

Note: When on ground the aircraft must squawk Mode C,in order to provide the altitude information to the Surveillance System.

IV. 1. Details of TORA for intersection departure taxiways:

Runway	Taxiway	TORA from Taxiway given in meters
19L	P	3451
	K	2749
	A	2126
	B	1578
01R	D	3555
	C	3174
	B	2054
19R	N	3216
	A	2698
	B	2255
	E	1518
	C	1135
01L	A	3257
	J	3149
	H	2817
	D	2635
	C	2134

DETAILS OF LDA FOR ARRIVAL AIRCRAFT:

Runway	Taxiway	LDA from Taxiway given in meters
19L	D	3128
	C	2748
	R	2349
	R1	1692
	B	1627
	A	1079
01R	P	3451
	K	2749
	Q2	2155
	A	2126
	Q1	1685
	B	1578
19R	A	3169
	J	3061
	H	2729
	D	2547
	C	2046
	E	1663
	B	926
01L	A	2260
	B	1817
	E	1080

VECC AD 2.24 CHARTS RELATED TO AN AERODROME

1 Aerodrome Chart (23 January 2025)
2 Aircraft Parking / Docking Chart (29 October 2026)
3 Aircraft Parking / Docking Chart Annexure A (29 October 2026)
4 Aircraft Parking / Docking Chart Apron 4 (29 October 2026)
5 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 01L(02 OCT 2025)
6 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 01R(02 OCT 2025)
7 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 19L(02 OCT 2025)
8 Aerodrome Obstacle Chart Type-A (Operating Limitations) RWY 19R(02 OCT 2025)
9 Aerodrome Obstacle Chart Type-B (02 OCT 2025)
10 Standard Airport Taxi Route-ARR RWY-01L Route 01A3

11 Standard Airport Taxi Route-ARR RWY-01R Route 01A1

12 Standard Airport Taxi Route-ARR RWY-01R Route 01A2

13 Standard Airport Taxi Route-ARR RWY-19L Route 19A1-A2

14 Standard Airport Taxi Route-ARR RWY-019R Route 19A3

15 Standard Airport Taxi Route-DEP RWY-01L Route 01D2

16 Standard Airport Taxi Route-DEP RWY-01L Route 01D3

17 Standard Airport Taxi Route-DEP RWY-01R Route 01D1

18 Standard Airport Taxi Route-DEP RWY-19L Route 19D1

19 Standard Airport Taxi Route-DEP RWY-19L Route 19D2

20 Standard Airport Taxi Route-DEP RWY-19R Route 19D3

21 Take Off Run Available from different Entry TWYs RWY 19L (23 January 2025)

22 Take Off Run Available from different Entry TWYs RWY 01R (23 January 2025)

23 Take Off Run Available from different Entry TWYs RWY 19R (23 January 2025)

24 Take Off Run Available from different Entry TWYs RWY 01L (23 January 2025)

25 LDA from Different Exit TWYs RWY 19L (23 January 2025)

26 LDA from Different Exit TWYs RWY 01R (23 January 2025)

27 LDA from Different Exit TWYs RWY 19R (23 January 2025)

28 LDA from Different Exit TWYs RWY 32 (23 January 2025)

29 Instrument Approach Chart - ILS or LOC Procedure RWY 01R (6 AUG 2026)

30 Instrument Approach Chart - ILS (CAT-II/III) Procedure RWY 01R (06 AUG 2026)

31 Instrument Approach Chart - ILS (Z) (CAT-I) Procedure RWY 19L

32 Instrument Approach Chart - ILS or LOC Procedure RWY 19R (6 AUG 2026)

33 Instrument Approach Chart - VOR Procedure RWY 01R (06 AUG 2026)

34 Instrument Approach Chart - VOR Procedure RWY 19L (06 AUG 2026)

35 Instrument Approach Chart - VOR Procedure RWY 19R (06 AUG 2026)

36 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R/01L, AGODA, ONOTO & DUMKA, 2E/2F (08 August 2024)

37 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R, AGODA, ONOTO & DUMKA, 2E - Tabular Description (08 August 2024)

38 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01L, AGODA, ONOTO & DUMKA, 2F - Tabular Description (08 August 2024)

39 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R/01L, BEMAK, NOKAT, SUMAG, AVPOP & DOPID, 2E/2F (17 April 2025)

40 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R, BEMAK, NOKAT, SUMAG, AVPOP & DOPID, 2E - Tabular Description (08 August 2024)

41 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01L, BEMAK, NOKAT, SUMAG, AVPOP & DOPID, 2F - Tabular Description (08 August 2024)

42 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R/01L, BOKEV, KAKID & SURUP, 2E/2F (08 August 2024)

43 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R, BOKEV, KAKID & SURUP, 2E - Tabular Description (08 August 2024)

44 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01L, BOKEV, KAKID & SURUP, 2F - Tabular Description (08 August 2024)

45 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R/01L, TEPAL & SEGMO, 2E/2F (08 August 2024)

46 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R, TEPAL & SEGMO, 2E - Tabular Description (08 August 2024)

47 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01L, TEPAL & SEGMO, 2F - Tabular Description (08 August 2024)

48 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R/01L, JJS, 2E/2F (08 August 2024)

49 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 01R/01L, JJS, 2E/2F - Tabular Description (08 August 2024)

50 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L/19R, SEGMO, DUMKA, ONOTO & AGODA, 2C/2D (08 August 2024)

51 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L, SEGMO, DUMKA, ONOTO & AGODA, 2C - Tabular Description (08 August 2024)

52 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19R, SEGMO, DUMKA, ONOTO & AGODA, 2DC - Tabular Description (08 August 2024)

53 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L/19R, BEMAK, NOKAT & SUMAG, 2C/2D (08 August 2024)

54 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L, BEMAK, NOKAT & SUMAG, 2C - Tabular Description (08 August 2024)

55 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19R, BEMAK, NOKAT & SUMAG, 2D - Tabular Description (08 August 2024)

56 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L/19R, AVPOP & DOPID, 2C/2D (17 April 2025)

57 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L, AVPOP & DOPID, 2C - Tabular Description (08 August 2024)

58 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19R, AVPOP & DOPID, 2D - Tabular Description (08 August 2024)

59 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L/19R, TEPAL & JJS, 2C/2D (08 August 2024)

60 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L, TEPAL & JJS, 2C - Tabular Description (08 August 2024)

61 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19R, TEPAL & JJS, 2D - Tabular Description (08 August 2024)

62 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L/19R, BOKEV, KAKID & SURUP, 2C/2D (08 August 2024)

63 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19L, BOKEV, KAKID & SURUP, 2C - Tabular Description (08 August 2024)

64 Standard Departure Chart – Instrument (SID), RNAV1 (GNSS), RWY 19R, BOKEV, KAKID & SURUP, 2D - Tabular Description (08 August 2024)

65 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R/01L, AGODA, 2G/2H (08 August 2024)

66 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R/01L, AGODA, 2G/2H - Tabular Description (08 August 2024)

67 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R/01L, BEMAK, NOKAT & SUMAG, 2G/2H (08 August 2024)

68 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R, BEMAK, NOKAT & SUMAG, 2G - Tabular Description (08 August 2024)

69 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01L, BEMAK, NOKAT & SUMAG, 2H - Tabular Description (08 August 2024)

70 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R/01L, JJS & TEPAL, 2G/2H (17 April 2025)

71 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R, JJS & TEPAL, 2G - Tabular Description (17 April 2025)

72 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01L, JJS & TEPAL, 2H - Tabular Description (17 April 2025)

73 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R/01L, KAKID,GOLAN, BOKEV, DOPID & AVPOP, 2G/2H (17 April 2025)

74 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R, KAKID, GOLAN, BOKEV, DOPID & AVPOP, 2G - Tabular Description (08 August 2024)

75 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01L, KAKID, GOLAN, BOKEV, DOPID & AVPOP, 2H - Tabular Description (08 August 2024)

76 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R/01L, ONOTO, DUMKA & SEGMO, 2G/2H (17 April 2025)

77 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01R, ONOTO, DUMKA & SEGMO, 2G - Tabular Description (17 April 2025)

78 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 01L, ONOTO, DUMKA & SEGMO, 2H - Tabular Description (17 April 2025)

79 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L/19R, JJS, TEPAL & SEGMO, 2A/2B (08 August 2024)

80 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L, JJS, TEPAL & SEGMO, 2A - Tabular Description (08 August 2024)

81 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19R, JJS, TEPAL & SEGMO, 2B - Tabular Description (08 August 2024)

82 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L/19R, AGODA, ONOTO & DUMKA, 2A/2B (08 August 2024)

83 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L, AGODA, ONOTO & DUMKA, 2A - Tabular Description (08 August 2024)

84 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19R, AGODA, ONOTO & DUMKA, 2B - Tabular Description (08 August 2024)

85 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L/19R, AVPOP, SUMAG, NOKAT & BEMAK, 2A/2B (08 August 2024)

86 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L, AVPOP, SUMAG, NOKAT & BEMAK, 2A - Tabular Description (08 August 2024)

87 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19R, AVPOP, SUMAG, NOKAT & BEMAK, 2B - Tabular Description (08 August 2024)

88 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L/19R, KAKID,GOLAN, BOKEV & DOPID, 2A/2B (17 April 2025)

89 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19L, KAKID,GOLAN, BOKEV & DOPID, 2A - Tabular Description (08 August 2024)

90 Standard Arrival Chart – Instrument (STAR), RNAV1 (GNSS), RWY 19R, KAKID,GOLAN, BOKEV & DOPID, 2B - Tabular Description (08 August 2024)

91 Instrument Approach Chart - RNP Y Procedure RWY 01L (16 May 2024)

92 Coding Table - RNP Y Procedure RWY 01L (16 May 2024)

93 Instrument Approach Chart - RNP Y Procedure RWY 01R (16 May 2024)

94 Coding Table - RNP Y Procedure RWY 01R (16 May 2024)

95 Instrument Approach Chart - RNP Y Procedure RWY 19L (16 May 2024)

96 Coding Table - RNP Y Procedure RWY 19L (16 May 2024)

97 Instrument Approach Chart - RNP Y Procedure RWY 19R (16 May 2024)

98 Coding Table - RNP Y Procedure RWY 19R (16 May 2024)

99 Instrument Approach Chart - ILS (CAT-II-III) Procedure RWY 19L

VECC AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	2	3
RNP Y RWY 01L	**OCA(H)=440(420) F**	**OBST: A) BUILDING LCA: 223802.71N 0882627.54E ELEV 95.14FT BUILDING LCA: 223801.84N 0882627.51E ELEV 94.49FT**